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Major human schistosome species express different glycans with immunological and diagnostic implications

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Laudine M. C. Petralia[1,2], Angela van Diepen[1], Tao Zhang[3], Anna O. Kildemoes[1], Tom Veldhuizen[1], D. Linh Nguyen[1], Cornelis H. Smit[1], Peter G. Kremsner[4,5], Manfred Wuhrer[3], Jeremy M. Foster[2] & Cornelis H. Hokke[1]

Schistosomes are parasitic worms responsible for devastating chronic disease including intestinal and urogenital pathologies. In-depth studies of the *Schistosoma mansoni* glycome have revealed complex, immunogenic and life stage-specific glycans, that are crucial in host-parasite interactions. Despite causing the majority of schistosome infections and specific associated pathology, the glycosylation of *S. haematobium* remains largely unstudied. Here, we characterize the glycan repertoire of *S. haematobium* using mass spectrometry-based approaches. We report substantial differences to *S. mansoni* glycans, most notably in the core structure, fucosylation patterns and glucuronic acid modifications of glycosphingolipid (GSL) glycans. Furthermore, IgG from *S. haematobium*-infected individuals preferentially bind the acidic GSL glycans compared to IgG from *S. mansoni*-infected individuals. Our results demonstrate that *S. haematobium* and *S. mansoni* are differentially glycosylated and that *S. haematobium* glycans are involved in parasite-host immunobiology. The specific, immunogenic *S. haematobium* glycans constitute targets for use in epidemiological studies and potential species-specific diagnostics.

Schistosomiasis is a widespread debilitating parasitic disease caused by trematodes of the genus *Schistosoma*. *S. mansoni* and *S. haematobium* are the most prevalent species causing human intestinal and urogenital schistosomiasis, respectively[1,2]. Disease control programs most commonly rely on mass administration of preventive chemotherapy, supported by safe water and hygiene (WASH) and educational initiatives[3,4]. There is no vaccine against schistosomiasis although novel (prophylactic) therapies as well as adequate diagnostics are deemed necessary to eliminate schistosomiasis as a global public health problem[5]. Thus, acquiring a solid fundamental knowledge of schistosome biology is critical to providing the field with the tools required to meet the 2030 elimination target.

Years of research into schistosome glycobiology, with an almost exclusive focus on *S. mansoni*, have unraveled structural, biological, and immunological features of the parasite's glycoconjugates. *S. mansoni* glycans are key players in host-parasite interactions and targets of the host's innate and adaptive immune system[6–8]. Extensive structural characterization has shed light on the large diversity of *N*-linked, *O*-linked and glycosphingolipid (GSL) glycans expressed by *S. mansoni*, that exhibit unique motifs such as dimeric α(1-2)-linked fucose (Fuc) residues and distinct sex- and life stage-specific features[9–11]. A strong and specific antibody response to a subset of *S. mansoni* glycans is raised by infected mammalian hosts both in natural and experimental infections[12–15]. These studies highlighted the

[1]Leiden University Center for Infectious Diseases, Leiden University Medical Center, Leiden, The Netherlands. [2]Biochemistry and Microbiology Division, New England Biolabs, Ipswich, MA, USA. [3]Center for Proteomics and Metabolomics, Leiden University Medical Center, Leiden, The Netherlands. [4]Centre de Recherches Médicales de Lambaréné, Lambaréné, Gabon. [5]Institut für Tropenmedizin, Universitätsklinikum Tübingen and Deutsches Zentrum für Infektionsforschung, Partner Site Tübingen, Tübingen, Germany. e-mail: c.h.hokke@lumc.nl

potential of schistosome glycan antigens as targets for diagnostic serology and as vaccine candidates[16–19]. Such structural or functional glycomics studies have not been performed for *S. haematobium* despite the specific and highly significant public health impact associated with urogenital schistosomiasis. *S. haematobium* infections can lead to disease such as male and female genital schistosomiasis[20,21] and bladder cancer[22]. Furthermore, *S. haematobium* is the human schistosome species most frequently involved in the worrisome zoonotic hybridizations that are increasingly observed with cattle schistosome species[23]. Compared to *S. mansoni*, the *S. haematobium* life cycle is more challenging to maintain in the laboratory resulting in lesser availability of species-specific material for research, including glycomics. Thus, general knowledge of *S. haematobium* glycobiology is limited to a few glycomic traits shared between *Schistosoma* species[24], including the excretion of the Circulating Cathodic Antigen (CCA) and Circulating Anodic Antigen (CAA); two diagnostic carbohydrate antigens[25,26]. Although many (putative) glycosylation-related genes of *S. mansoni*[27] have orthologues in *S. haematobium*[28], differences between the two species are expected in view of their specific biology.

Here, we studied the lipid-linked GSL glycans and the protein-linked *N*- and *O*-glycans expressed by *S. haematobium* cercariae, adult worms and eggs in detail. We combined mass spectrometry (MS)-based approaches with enzymatic and chemical glycan sequencing to structurally define glycans in these three key life-stages of the parasite. We found that the *S. haematobium* glycome comprises many glycan elements shared with *S. mansoni* but expressed with major quantitative differences. Importantly, the identified *S. haematobium* GSL glycans also harbor species-specific structural features including a trihexosyl GSL core, cercarial branched structures with β-linked galactose (Gal), fucosylation patterns and glucuronic acid (GlcA) modifications in the eggs. A microarray with a specific focus on the differentially expressed native *S. haematobium* and *S. mansoni* glycans was constructed and screened with sera from schistosome-infected individuals, which revealed differential binding of IgG from *S. mansoni* and *S. haematobium* infected individuals to acidic GSL glycans in particular.

This study provides a detailed analysis of the hitherto unknown glycome and glycan antigen repertoire of *S. haematobium*. Our findings indicate that many *S. haematobium* glycan antigens, including some that are unique to the species, are immunogenic. This suggests that S. *haematobium*-specific glycans play a role in *S. haematobium*-host interactions that could be exploited as species-specific serology targets for epidemiological studies.

Results

I] Structural analysis of *S. haematobium* GSL, *N*- and *O*-linked glycans reveals major structural differences compared to *S. mansoni*

Lipid- and protein-linked glycans from *S. haematobium* cercariae, adult worms and eggs were released enzymatically using endoglycoceramidase II (rEGCase II) for the GSL glycans and *N*-glycosidases (PNGases) for the *N*-glycans, or chemically, via reductive β-elimination for the *O*-glycans. Released glycans were analyzed by Matrix Assisted Laser Desorption Ionization Time-of-Flight (MALDI-TOF)-MS, revealing complex spectra with many molecular ions (Fig. S1 and Supplementary data 1), similar to previous observations for *S. mansoni*[9]. To tackle this complexity and allow efficient comparison between schistosome species, we focused on glycans yielding ions with signal intensities >1% of the total MALDI-TOF-MS spectrum intensity and presence in replicate measurements (Supplementary data 2, GSL glycans; Supplementary data 3, *N*-glycans, and Supplementary data 4, *O*-glycans). Glycan species are referred to throughout the manuscript using their theoretical masses in negative-ion mode for GSL and *N*-glycans ($[M\text{-}H]^-$) and in positive-ion mode for *O*-glycans ($[M+Na]^+$).

S. haematobium GSL glycans are built on a trihexosyl core extended with oligomeric chains of β-linked N-acetylhexosamine (HexNAc) residues

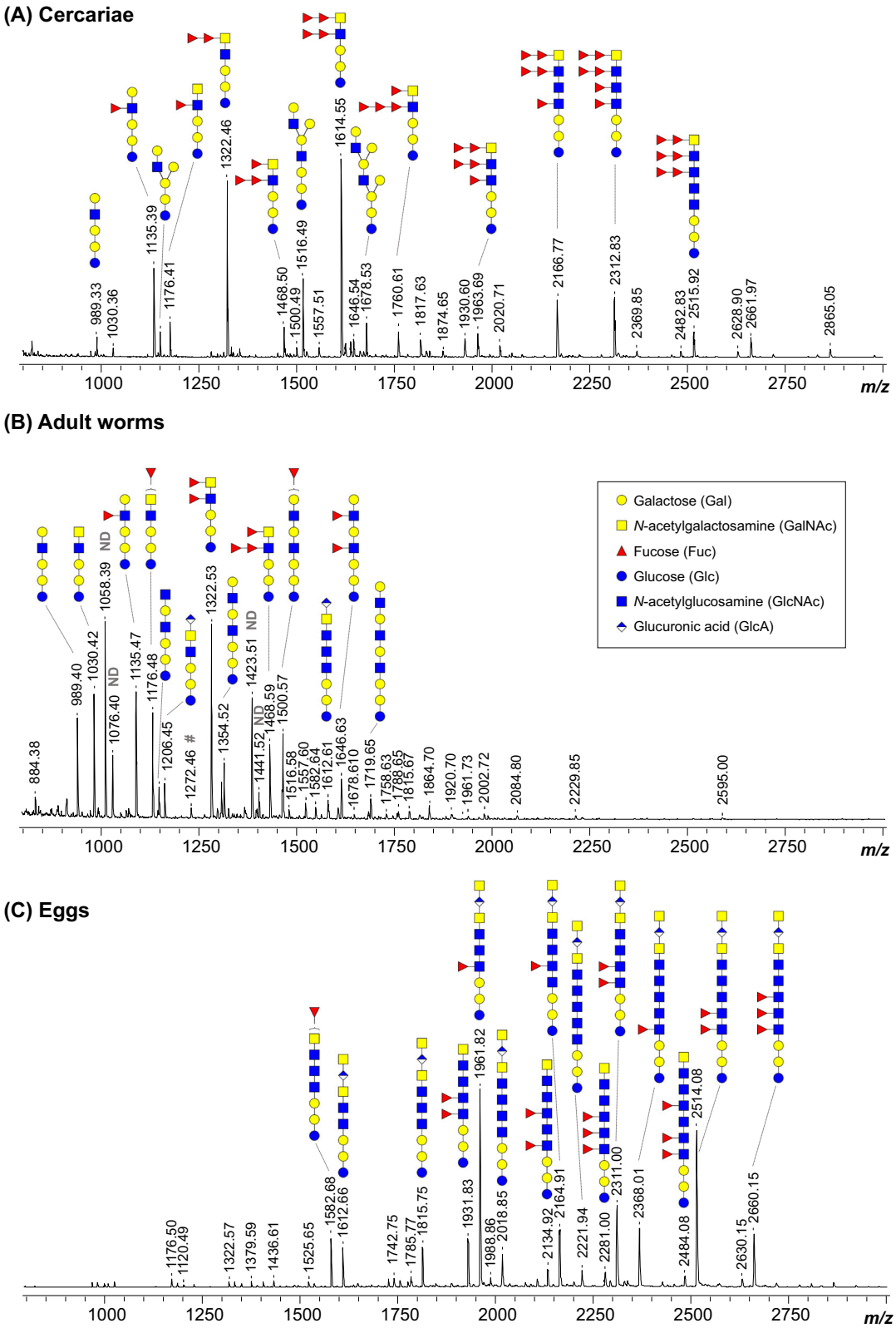
MALDI-TOF-MS analysis of 2-aminobenzoic acid (AA)-labeled *S. haematobium* GSL glycans (Fig. 1) showed striking differences to the previously described *S. mansoni* GSL glycan profiles[9]. For all three life stages examined, the *S. haematobium* GSL glycan backbone invariably contained a minimum of three hexose (Hex, H) residues (Supplementary Data 1-2). β-N-acetylhexosaminidase (β-HexNAcase, Fig. 2) could cleave N-acetylhexosamine (HexNAc, N) residues from GSL glycans of composition $H_3N_{2\text{-}6}$-AA (*m/z* 1030.37; 1233.45; 1436.53; 1639.61; 1842.63) that were either present in the native samples or obtained through defucosylation of structures of composition $H_3N_{2\text{-}6}F_{1\text{-}6}$-AA using hydrofluoric acid (HF) (Fig. S2.A–C). Resulting structures with $m/z < 700$ could not be detected by MALDI-TOF-MS, implying that the three hexoses are located at the reducing end of the glycans. Y-type fragment-ions corresponding to Y_2-H_2-AA and Y_3-H_3-AA (*m/z* 462.16; 624.21) were visible in MALDI-TOF-MS/MS fragmentation spectra of GSL glycans indicating that the three Hex residues are present in a linear stretch at the reducing end of the GSL backbone (Fig. S3.A–C). We used an ultra-high performance liquid chromatography (UHPLC) platform, optimized for the detection of short oligosaccharides[29], to further analyze the core trisaccharide that was obtained from exoglycosidase digestion of HF-treated *S. haematobium* egg GSL glycans (Fig. S4). While β1-3,4 galactosidase had no effect on the structure, digestion with α-galactosidases—both with α1-3,6 and α1-3,4,6 galactosidases—resulted in a disaccharide product, that could be further digested using β1-4 and β1-3,4 galactosidase (Fig. 2). These results indicated that *S. haematobium* GSL glycans are built on a trihexosyl core constituted of two galactoses most likely extending a ceramide (Cer)-linked glucose (Glc) and forming the isoglobo-series core Galα1-3Galβ1-4Glcβ1-Cer that has also been reported in the GSL glycans of the liver fluke *Fasciola hepatica*[30,31]. The trihexosyl core could also be detected using porous graphitized carbon nano-liquid chromatography coupled to tandem mass spectrometry (PGC-nano-LC-MS/MS, Fig. S5). Corroborating the UHPLC analysis, the *S. haematobium* GSL glycan core exhibited a different elution pattern in the PGC-nano-LC-MS analysis compared to the synthetic globotriose (Gb3) standard Galα1-4Galβ1-4Glc, but its elution profile was consistent with that of the isoglobotriose (iGb3) standard Galα1-3Galβ1-4Glc (Fig. S5.A). The occurrence of this trihexosyl core structure in *S. haematobium* greatly contrasts with *S. mansoni* GSL glycans where the ceramide-linked glucose (Glc) is extended with β1-4-linked N-acetylgalactosamine (GalNAc)[32,33].

The MALDI-TOF-MS/MS analysis (Fig. S3.A–C) and β-HexNAcase digestions of HF-treated structures (Fig. S2.A–C) mentioned previously showed that stretches of β-linked HexNAc residues extend the trihexosyl core in most *S. haematobium* GSL glycans. Resistance to treatment with *Streptococcus pneumoniae* β-N-acetylglucosaminidase (β-GlcNAcase) indicated that β1-4-linked GalNAc is capping the structures with $H_3N_{2\text{-}6}F_{0\text{-}6}$ compositions (Fig. S2.A), as β-GlcNAcase lacks activity towards β1-4 linked GalNAc (Fig. 2). This is similar to previous observations in *S. mansoni* where HexNAc stretches of the GSL backbones were shown by monosaccharide analysis to consist predominantly of N-acetylglucosamine (GlcNAc) with the exception of the terminal and core-GalNAc residues[32,34–36]. Thus, unless determined otherwise, we assumed the backbone HexNAc to be primarily constituted of β-linked GlcNAc, by analogy (Fig. 3).

Acidic GSL glycans in *S. haematobium*: identification of terminal glucuronic acid (GlcA) in adult worms and abundant subterminal GlcA-containing glycans in eggs

Several ions detected in the MALDI-TOF-MS spectra of *S. haematobium* worm and egg GSL glycans could not be assigned to glycan compositions containing solely monosaccharide constituents detected

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previously in *S. mansoni* GSL glycans[9]. PGC-nano-LC-MS/MS analysis of the ion with *m/z* 1087.37 detected in adult worms showed a clear loss of 176.03 Da directly from the parent ion resulting in an ion with *m/z* 911.34 (Y_5-H_3N_2) (Fig. S5.B). A mass increment of 176 Da matches the mass of either a methylated hexose or a hexuronic acid (HexA, A) two types of residues that have been observed in helminths and mollusk glycans, including in *Biomphalaria* snails[37,38]. Efficient ionization in negative-ion mode indicated that the ion species with *m/z* 1206.41 and *m/z* 1612.56 contained HexA, with compositions of $H_3N_2A_1$ and $H_3N_4A_1$-AA, respectively. In line with the fragmentation pattern, β-HexNAcase had no effect on these structures, while both could be digested using β-glucuronidase (Fig. S2.B), showing that a terminal β-linked GlcA is capping these adult worm GSL glycans (Fig. 1B and Supplementary data 2). The β-glucuronidase digestion product (*m/z* 1030.37 (H_3N_2-

Fig. 1 | MALDI-TOF-MS of S. haematobium GSL glycans. GSL glycans were released from *S. haematobium* cercariae (**A**), adult worm (**B**) and egg (**C**) glycolipids using rEGCase II prior to labeling with AA. Labeled GSL glycans were analyzed using MALDI-TOF-MS in negative-ion reflectron mode. Monoisotopic masses of measured signals are indicated ([M-H]$^-$). Proposed glycan structures are depicted using the Consortium for Functional Glycomics (CFG) nomenclature (see symbol key insert) for ions with signal-to-noise ratios > 4, intensities above 1% of the total spectrum intensity and detected in both biological replicates (Supplementary data 1). Compositions and structures were deduced using a panel of glycan sequencing techniques in combination with MALDI-TOF-MS/MS fragmentation, PGC-nano-LC-MS(/MS) analysis when applicable (Supplementary data 2) and information from literature. ND (non-determined) indicates *m/z* to which a glycan structure could not be assigned applying this rationale. We assumed the backbone HexNAc to be GlcNAc, based on analogy with *S. mansoni* for which HexNAc stretches of the GSL backbones have been shown by monosaccharide analysis to be constituted predominantly of GlcNAc with the exception of the terminal and core-GalNAc residues[32,34–36]. Note that we identified instances where the HexNAc preceding the GlcA residue could be either GlcNAc or GalNAc (Supplementary data 2). GalNAc-containing isomers are represented in this figure.

Substrate	Enzyme/reagent name	Source (cat. #)
α1-3/4/6 — R	**α1-3,4,6 galactosidase**	Green coffee bean, expr. in *Escherichia coli* (NEB, #P0747)
α1-3/6 — R	**α1-3,6 galactosidase**	*Xanthomonas manihotis*, expr. in *Escherichia coli* (NEB, #P0731)
α1-2/3/6 — R	**α1-2,3,6 mannosidase**	*Canavalia ensiformis* (Jack Bean), expr. in *Pichia pastoris* (NEB, #P0768)
β1-4 — R	**β1-4 galactosidase**	*Streptococcus pneumoniae*, expr. in *Escherichia coli* (NEB, #P0745)
β1-3/4 — R	**β1-3,4 galactosidase**	*Bovine testis*, expr. in *Pichia pastoris* (NEB, #P0746)
β1-2/3/4/6 — R	**β-N-acetylglucosaminidase (β-GlcNAcase)**	*Streptococcus pneumoniae*, expr. in *Escherichia coli* (NEB, #P0744)
β1-3/4/6 — R; β1-4 — R	**β-N-acetylhexosaminidase (β-HexNAcase)**	*Streptomyces plicatus*, expr. in *Escherichia coli* (NEB, #P0721)
β — R	**β-glucuronidase**	*Patella vulgata* expr. in *Patella vulgata* (Millipore Sigma, #SRE0093)
α1-2/3/4 — R	**Hydrofluoric acid (HF)**	N/A (Millipore Sigma, #695068)

Galactose (Gal); *N*-acetylgalactosamine (GalNAc); *N*-acetylglucosamine (GlcNAc); Glucuronic acid (GlcA); Fucose (Fuc); Mannose (Man)

Fig. 2 | Substrate specificities of the exoglycosidases and hydrofluoric acid (HF) used for glycan sequencing. HF and β-glucuronidase were obtained from Millipore Sigma, while all other enzymes were from New England Biolabs (NEB). The enzyme source and expression (expr.) organisms are specified together with catalog numbers (cat. #).

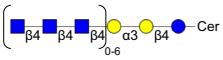
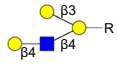
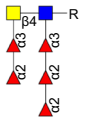
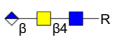
AA)) was then sensitive to β-HexNAcase but was resistant to β-GlcNAcase (Fig. S2.B) showing that GlcA was linked to GalNAc (Fig. 2).

HexA-containing GSL glycans of composition of $H_3N_{2\text{-}7}A_1F_{0\text{-}3}$ constituted a major subset of *S. haematobium* egg GSL glycans (Fig. 1C and Supplementary data 2). Unlike in adult worms, MS/MS fragmentation patterns suggested a subterminal position of the HexA residue, with the visible loss of a HexNAc (203.08 Da) from parent ions immediately followed by the loss of HexA (176.03 Da) (Fig. S3.C, MALDI-TOF-MS/MS and Fig. S5.C, PGC-nano-LC-MS/MS). Further glycan sequencing work showed that β-HexNAcase could cleave a single HexNAc from these HexA-containing glycans (Fig. S2.C). Upon removal of the capping HexNAcs, these egg GSL glycans became sensitive to β-glucuronidase, confirming the presence of subterminal β-linked GlcA. The products of β-glucuronidase-treatment could then be further digested with β-HexNAcase to structures of composition $H_3N_{2\text{-}4}F_{1\text{-}3}$. β-GlcNAcase was also effective on a substantial proportion of the β-glucuronidase digestion products (Fig. S2.C, Supplementary data 2), suggesting that the GlcA can be β-linked to either GalNAc or GlcNAc in the egg GSL glycan backbone (Fig. 2). For simplicity, we represented GalNAc-containing isomers in Figs. 1, 4, and S2–5.

Lower abundance of acidic GSL glycans in *S. mansoni* eggs compared to *S. haematobium*

Acidic GSL glycans have not been reported for *S. mansoni*[9], although preliminary data suggested that some *S. mansoni* egg GSL glycans contain a HexA residue[39]. To further investigate this and perform a direct comparison with *S. haematobium*, we conducted a parallel glycomic analysis of egg GSL glycans from both species (Fig. 4). In line with previous data, all 10 major signals in the *S. mansoni* egg profile were assigned to neutral structures of composition $H_1N_{3\text{-}7}F_{2\text{-}8}$, unlike in

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	Structure	Glycan symbol	Life stages		
			Cercariae	Worms	Eggs
Backbone	**Trihexosyl core +/- GlcNAc stretch** (GlcNAcβ1-4)$_n$Galα1-3Galβ1-4Glc-Cer		Present in all structures & life stages		
Terminal motifs	**Branched β1-3 Gal** Galβ1-4GlcNAcβ1-4(Galβ1-3)Gal		1151.40 1516.53 1678.58		
	LN, LacNAc Galβ1-4GlcNAc		989.35	989.35 1354.48 1500.54 1719.61	
	LeX, Lewis X Galβ1-4(Fucα1-3)GlcNAc		1135.40 1646.59	1135.40 1500.54 1646.59	
	DF-LDN-TF Fucα1-2Fucα1-3GalNAcβ1-4(Fucα1-2Fucα1-2Fucα1-3)GlcNAc		1760.66		
	DF-LDN-DF Fucα1-2Fucα1-3GalNAcβ1-4(Fucα1-2Fucα1-3)GlcNAc		1614.60 1963.74 2166.82 2312.88 2515.95		
	DF-LDN Fucα1-2Fucα1-3GalNAcβ1-4GlcNAc		1322.49		
	F-LDN-DF Fucα1-3GalNAcβ1-4(Fucα1-2Fucα1-3)GlcNAc		1468.55	1468.55	
	F-LDN-F Fucα1-3GalNAcβ1-4(Fucα1-3)GlcNAc			1322.49	
	LDN-F GalNAcβ1-4(Fucα1-3)GlcNAc		1176.43	1176.43	1176.43 1582.59
	F-LDN Fucα1-3GalNAcβ1-4GlcNAc			1176.43	1176.43
	LDN, LacDiNAc GalNAcβ1-4GlcNAc			1030.37	1030.37 1436.53 1931.72 2134.80 2280.86 2483.94
	Subterminal GlcA GalNAcβ1-4GlcAβ-HexNAc				1612.56; 1815.64; 1961.70; 2018.72; 2164.78; 2221.80; 2310.84; 2368.86; 2513.92; 2659.98
	Terminal GlcA GlcAβ-GalNAcβ1-4GlcNAc			1206.41 1612.56	

Fucose (Fuc) · Glucuronic acid (GlcA) · Galactose (Gal) · Glucose (Glc) · *N*-acetylgalactosamine (GalNAc) · *N*-acetylglucosamine (GlcNAc) · GalNAc or GlcNAc

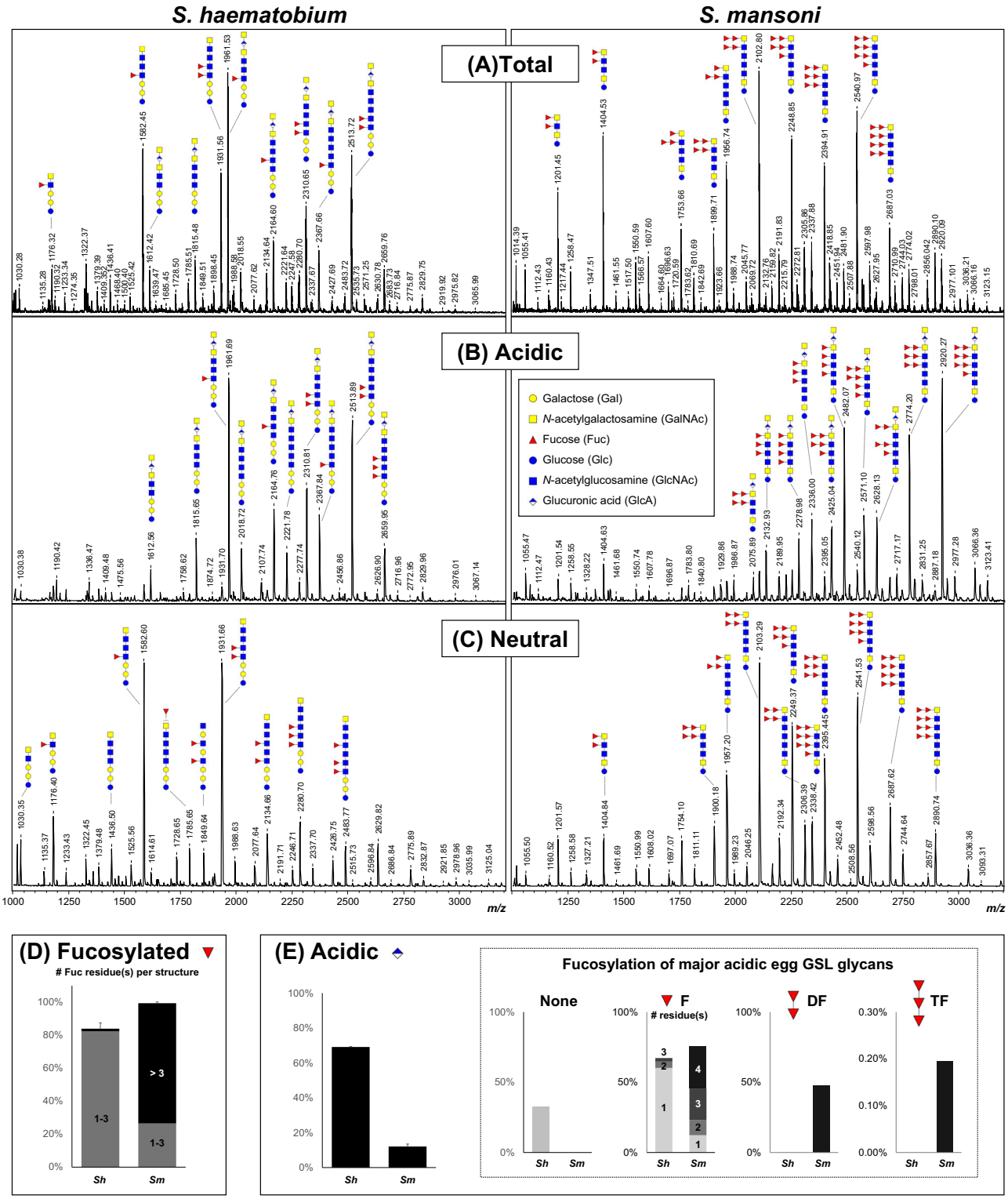
Motif present in life stage in listed glycan structures (*m/z* of AA-labeled GSL glycans in negative ion mode, see **Table S2**)

Motif not detected in life stage

Fig. 3 | Distribution of structural motifs in *S. haematobium* GSL glycans across life stages. Presence of specific glycan motifs in *S. haematobium* life stages as determined using MALDI-TOF-MS/MS (Fig. S3), UHPLC (Fig. S4), and PGC-nano-LC-MS/MS (Fig. S5) in combination with glycan sequencing (Fig. S2). Glycan motifs are depicted using the CFG nomenclature.

S. haematobium, where 7 out of the 10 major ions contain HexA (Fig. 4A, Supplementary data 2). When acidic and non-acidic glycans were separately eluted and subjected to MALDI-TOF-MS analysis (Fig. 4B, C), the HexA-containing glycans described above were recovered in the acidic GSL glycan fraction from *S. haematobium* eggs, while the *m/z* of the main ions detected in the neutral fraction corresponded to structures of compositions $H_3N_{2\text{-}7}F_{0\text{-}3}$ (Supplementary data 2). For *S. mansoni* eggs, the *m/z* values observed in the neutral fraction accurately matched the GSL glycans already defined[9], while the acidic profile revealed a substantial number of previously

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uncharacterized molecular ions with compositions including HexA, that constitute a minor subset of the total *S. mansoni* GSL glycan pool (Fig. 4). Structural characterization indicated a similar subterminal context for the HexA residue in the backbone of *S. mansoni* egg GSL glycans (Fig. 4 and Supplementary data 2). Digestion of a single HexNAc was obtained by applying β-HexNAcase to the HF-defucosylated glycans (Fig. S2.E) and the invariable loss of one HexNAc followed by one HexA from parent ions was visible in MALDI-TOF-MS/MS (Fig. S3.D) and PGC-nano-LC-MS/MS analysis (Fig. S5.E-F). In both *S. haematobium* and *S. mansoni* eggs, the HexNAc residue capping acidic GSL glycans was never fucosylated (Fig. S3.C-D and Fig. S5.C, E-F), and comparison between β-HexNAcase and β-GlcNAcase activities (Fig. 2) suggested the majority of these capping HexNAc residues to be a β1-4-linked GalNAc (Fig. S2.C, E).

Differential fucosylation of *S. haematobium* and *S. mansoni* egg GSL glycans

Fucosylation levels varied substantially between *S. haematobium* and *S. mansoni* egg GSL glycans. While the main GSL glycans of *S. haematobium* have a maximum of three Fuc residues and several acidic

Fig. 4 | Comparison of acidic and neutral GSL glycans of *S. haematobium* and *S. mansoni* eggs using MALDI-TOF-MS and PGC-nano-LC-MS. GSL glycans were released from the eggs of *S. haematobium* and *S. mansoni* using rEGCase II. GSL glycans were purified using carbon-SPE purification (see Methods section) and either eluted as a single, total fraction (**A**) or as separate acidic (**B**) and neutral (**C**) glycan fractions, as indicated. Glycans were labeled with AA and analyzed using MALDI-TOF-MS in negative-ion reflectron mode (**A**–**C**). Raw data can be found in Supplementary data 1. Monoisotopic masses of measured signals are indicated ($[M-H]^-$) and proposed glycan structures for the ten most abundant ions based on percentage of total spectrum intensity are depicted using the CFG nomenclature (see symbol key insert). Compositions and structures were deduced using a panel of glycan sequencing techniques in combination with MS/MS fragmentation and information from literature. When isomeric structures were detected and characterized using PGC-nano-LC-MS (see Supplementary data 2), the most abundant isomer is represented. As in Fig. 1, in instances where both GlcNAc and GalNAc were found preceding the GlcA residue, the GalNAc-containing isomer is represented. MALDI-TOF-MS intensity signals were used to estimate the proportion of fucosylated (Fuc-containing) (**D**) and acidic (GlcA-containing) (**E**) GSL glycans in *S. haematobium* (*Sh*) and *S. mansoni* (*Sm*) eggs. Relative abundance is displayed as mean of summed peak intensities, with standard deviation between duplicates as error bars. In addition, the type and proportion of fucosylated motifs observed on the main acidic GSL glycans analyzed by PGC-nano-LC-MS (Fig. S5) was calculated using areas of extracted ion chromatograms (EICs) (E). Proportion of glycans found to be not fucosylated (none), carrying single Fuc residues (F, with number of Fuc residues per structure indicated), di-fucosyl (DF, Fucα1-2Fucα1-3 DF) and tri-fucosyl (TF, Fucα1-2Fucα1-2Fucα1-3) motifs are represented. The raw data used to generate the graphs are provided as a Source Data file.

GSL glycans are not fucosylated, the major structures identified in *S. mansoni* comprised three to eight fucoses (Fig. 4 and Supplementary data 2). Both the acidic and neutral GSL glycans of *S. haematobium* eggs were sensitive to β-HexNAcase, that could cleave up to four HexNAc residues from the neutral ones (Fig. S2.C). Resistance to treatment with β-GlcNAcase indicated that β1-4-linked GalNAc is capping neutral structures with $H_3N_{2\text{-}7}F_{0\text{-}3}$ compositions, forming GalNAcβ1-4GlcNAc (LacDiNAc, LDN) motifs, as in *S. mansoni*[9,33,36]. In *S. mansoni* egg GSL glycans, however, these terminal LDN motifs were found to be fucosylated, as previously reported[9]. Similarly, in the acidic fraction, HF-defucosylation was necessary for β-HexNAcase to act (Fig. S2.E) due to the presence of multiple Fuc substituents near the non-reducing end of the GSL glycan backbone. Thus, Fuc residues are not only more abundant in *S. mansoni* egg GSL glycans, but their position on the GSL glycan backbone constitutes another major difference between species.

Resolving the structure of such (multi)fucosylated glycans by MALDI-TOF-MS/MS is notoriously challenging[33,40], but is enabled by the high isomer separation power of PGC chromatography. Using PGC-nano-LC-MS/MS, we detected several isomeric structures resulting from variable Fuc distribution among the acidic GSL glycans of *S. mansoni* (Fig. S5.F). We also found the HexNAc residue directly preceding HexA to be invariably fucosylated, and often with Fuc residues forming the di-fucosyl motifs (Fucα1-2Fucα1-3, DF). In contrast, DF elements were absent in *S. haematobium* egg GSL glycans for which HexNAc residues were found either unsubstituted or substituted only with a single Fuc (Fig. S5.C-D). When present, Fuc was found preferentially attached to the HexNAc residue(s) close to the trihexosyl core. In the acidic GSL glycans, this left the HexA preceded by a chain of three or more unsubstituted HexNAc residues. Fewer isomeric structures were identified in the GSL glycans of *S. haematobium* than in *S. mansoni* egg GSL glycans, which is a direct consequence of the lower degree of fucosylation in *S. haematobium* (Fig. 4). For *S. mansoni*, the abundance of highly fucosylated GSL structures was shown to increase with egg maturity[9]. To investigate a potential change in glycosylation during egg development in *S. haematobium*, we released and analyzed GSL glycans from separated immature and mature eggs. The resulting MALDI-TOF-MS spectra were virtually identical (Fig. S2.D), suggesting that, during maturation, the *S. haematobium* eggs do not develop the highly fucosylated motifs ubiquitous in the mature eggs of *S. mansoni*.

Branched β-Gal in *S. haematobium* cercarial GSL glycans

Several non-fucosylated cercarial GSL glycans appeared to have $H_{4\text{-}7}N_{1\text{-}2}$ compositions, containing hexoses in addition to the trihexosyl core (Figs. 1, 3, Supplementary data 2). These glycans were terminated with β−1-4 linked Gal as shown by sensitivity to β−1-4 galactosidase (Fig. S2.A). MALDI-TOF-MS/MS analysis of the β1-4 galactosidase digestion products at *m/z* 1354.48 (H_5N_2-AA) and 1516.53 (H_6N_2-AA) (Fig. S3.A) each showed a loss of both one Hex (fragment ions with *m/z* 1192.43 and 1354.48) and one HexNAc residue (fragment ions with *m/z* 1151.40 and 1313.45), suggesting these structures to be branched, an unusual trait in the typically linear GSL glycans. Further digestion with β-GlcNAcase, indicated the branched HexNAc to be β-linked GlcNAc while sensitivity to β1-3,4 galactosidase but not with β1-4 galactosidase (Fig. S2.A, Supplementary data 2) indicated the branched Hex to be a β1-3 Gal, in contrast to the terminal β1-4 Gal (Fig. 2). These glycans did not resemble any of those previously described for *S. mansoni*.

Life stage-specific expression of (fucosylated) terminal LDN and LacNAc (LN) motifs in *S. haematobium* GSL glycans

Fucosylation, including α1-2 linked Fuc residues forming DF or trifucosyl motifs (Fucα1-2Fucα1-2Fucα1-3, TF) is a unique and antigenic feature of *Schistosoma* glycans[13,41,42]. DF and TF motifs, which are particularly abundant in *S. mansoni* egg GSL glycans were not detected in the eggs of *S. haematobium* (Fig. 4) but were found in the GSL glycans of *S. haematobium* cercariae and worms (Fig. 1A, B), where they form fucosylated terminal LDN motifs (Fig. 3). In addition, terminal LacNAc (LN, Galβ1-4GlcNAc) and Lewis X (LeX, Galβ1-4(Fucα1-3)GlcNAc) motifs were observed in cercarial and adult worm GSL glycans (Figs. 1, 3, Supplementary data 2) as determined using MALDI-TOF-MS/MS analysis (Fig. S3.A-B) and sensitivity to HF treatment and to a combination of β1-4 galactosidase and β-GlcNAcase (Fig. S2.A-B). LN and LeX termination were found to be major features of *S. haematobium* adult worm GSL glycans, together with fucosylated LDN and the GlcA-containing motifs described above (Figs. 1, 3). This diversity of motifs contrasts with *S. mansoni* adult worms GSL glycans, largely terminated by (multi)fucosylated LDN motifs[9]. In addition, we did not find any evidence of pseudo-Lewis Y (Fucα1-3Galβ1-4(Fucα1-3)GlcNAc) in *S. haematobium* GSL glycans, while it is the most prominent terminal motif in *S. mansoni* cercarial GSL glycans[9].

N- and *O*-linked glycans of *S. haematobium* show quantitative differences to those of *S. mansoni*

N-linked glycans were released by sequential use of PNGase F and PNGase A and structurally characterized using glycan sequencing approaches (Fig. S6). Many of the features previously reported for *S. mansoni*[9] were also detected in *S. haematobium* *N*-glycans, including presence of terminal LN, LeX and (fucosylated) LDN motifs as well as the non-mammalian core-modifications β1-2 linked xylose and α1-3 linked core-Fuc (Fig. 5 and Supplementary data 3). Stage-specific expression of many glycan motifs was observed. *N*-glycans carrying α1-3 core-fucose, released with PNGase A, were exclusively detected in the mature eggs (Fig. 5D, Fig. S6.D-E). In addition, these glycans were found to include high molecular weight structures (> 2500 Da), most likely containing LN repeats based on their monosaccharide composition (Fig. S6.G). Interestingly, the MALDI-TOF-MS profiles of glycans released from immature and mature *S. haematobium* eggs by PNGase F were virtually identical (Fig. S6.F), which differs from the increase in

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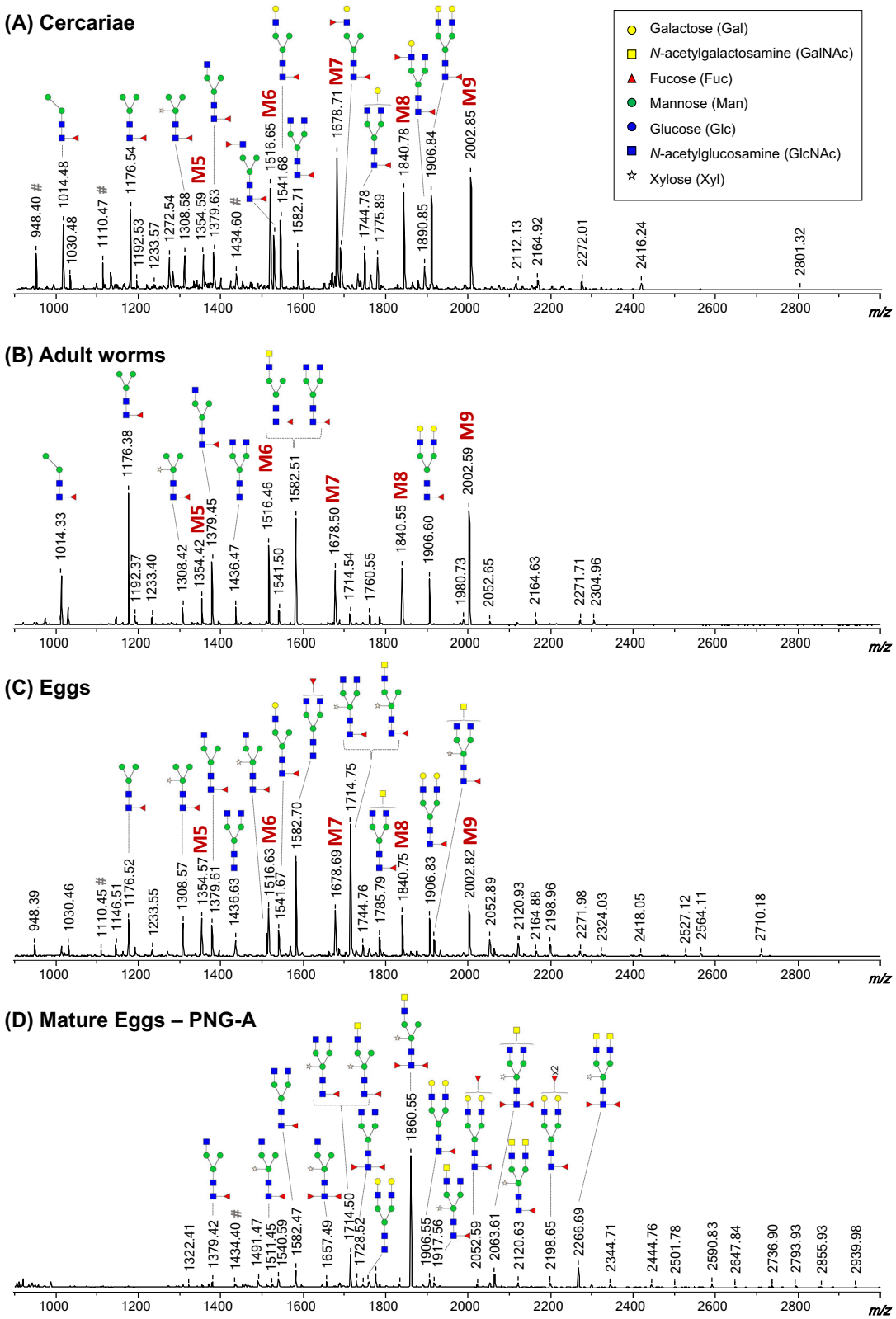


Fig. 5 | MALDI-TOF-MS of *S. haematobium* *N*-glycans. *N*-glycans were released from *S. haematobium* cercariae (**A**), adult worm (**B**) and egg (**C**, **D**) glycoproteins using PNGase F (**A**–**C**) and PNGase A, sequentially (D). This analysis was performed on a mixture of eggs at different stages of maturity (**C**) and on isolated immature/mature eggs (**D**). Released *N*-glycans were subsequently labeled with AA and analyzed using MALDI-TOF-MS in negative-ion reflectron mode. PNGase A-specific *N*-glycans α1-3 core-fucosylated, and α1-3/α1-6 core-difucosylated were observed in the mature eggs, as shown in panel **D**, but not in the immature eggs (data not shown). PNGase F profiles of immature eggs and mature eggs are available in Fig. S6.F and raw data can be found in Supplementary data 1. Monoisotopic masses of measured signals are indicated ($[M\text{-}H]^-$). Proposed glycan structures are depicted using the CFG nomenclature (see symbol key inset) for ions with signal-to-noise ratios > 4, intensities above 1% of total spectrum intensity and presence in both biological replicates. Known non-glycan signals are indicated using the # symbol. M5 to M9 are used to label oligomannosidic *N*-glycans with 5–9 mannose residues. Compositions and structures were deduced using a panel of glycan sequencing techniques in combination with MALDI-TOF-MS/MS fragmentation and information from the literature.

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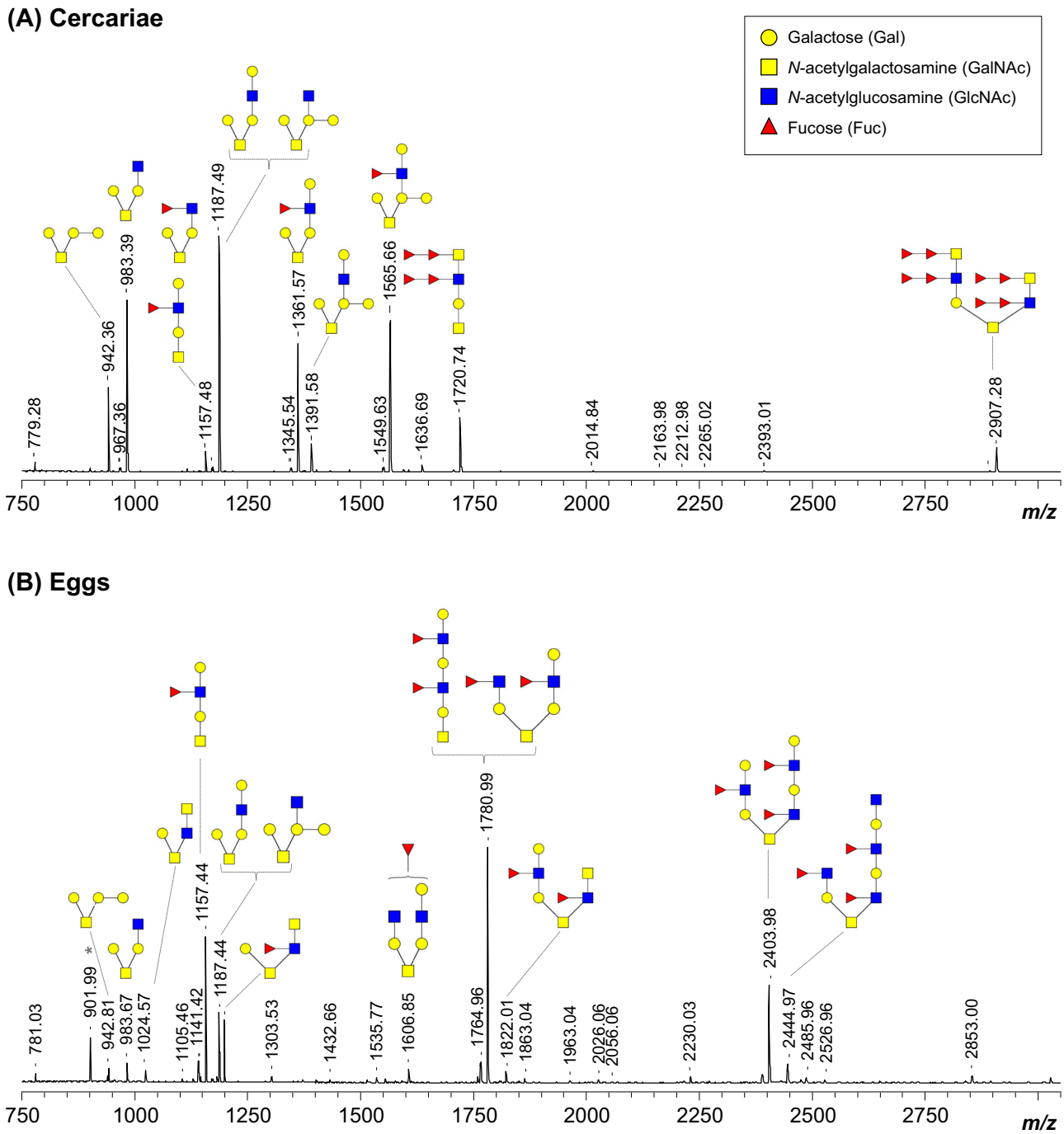


Fig. 6 | MALDI-TOF-MS of *S. haematobium* *O*-glycans. *O*-glycans were released by reductive β-elimination from *S. haematobium* glycoproteins and permethylated prior to MALDI-TOF-MS analysis. MALDI-TOF-MS profiles obtained for cercariae (**A**) and eggs (**B**) are shown. Spectra were recorded in positive-ion reflectron mode and raw data can be found in Supplementary data 1. Monoisotopic masses of measured signals are indicated ($[M+Na]^+$). Proposed glycan structures are depicted using the CFG nomenclature (see symbol key inset) for ions with signal-to-noise ratios >4, intensities above 1% of the total spectrum intensity and presence in both technical replicates. Structures were characterized by MALDI-TOF-MS/MS fragmentation supplemented with information from the literature.

complex, highly fucosylated *N*-glycans observed in *S. mansoni* during egg maturation[9]. Certain features of the evolution of *N*-glycan expression throughout the parasite life cycle appeared common to *S. haematobium* and *S. mansoni* such as the shift from cercarial terminal LN/LeX to LDN in adult worms and eggs[9]. However, many qualitative differences in the distribution of the glycan motifs were also observed between both species (Fig. S7) particularly for cercarial *N*-glycans. Generally, we found that *S. haematobium* *N*-glycans were shorter, less complex, and less fucosylated than those of *S. mansoni*, following the trend shown for the GSL glycans.

Following reductive β-elimination, released *O*-glycans were permethylated and analyzed by positive-ion mode MALDI-TOF-MS (Fig. 6). Glycan-derived signals were detected in each of the life stages, but with very low intensity for the adult worms, similar to *S. mansoni*[9], preventing further interpretation. Structural characterization of *S. haematobium* cercarial and egg *O*-glycans was performed using MALDI-TOF-MS/MS (Fig. S8) and indicated that this class of glycans is built either on the conventional *O*-glycan core-2 of composition H_1N_2 (Galβ1–3(GlcNAcβ1–6)GalNAc), or on a H_2N_1 schistosome-specific core (Galβ1-3(Galβ1-6)GalNAc)[43]. The majority of the cercarial *O*-glycans

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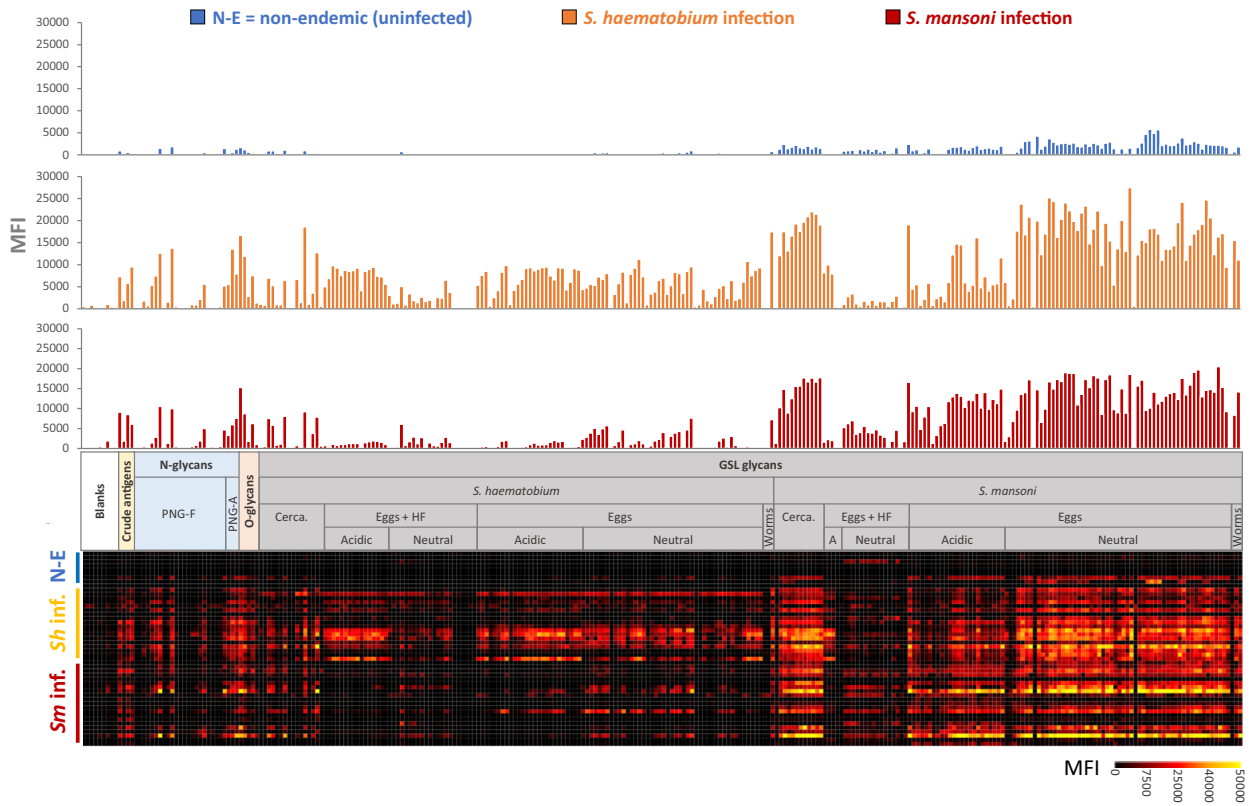


Fig. 7 | Serum IgG responses to *Schistosoma* glycans. Microarrays were incubated with sera from non-endemic individuals ($n = 9$) and with sera from children infected with *S. haematobium* ($n = 18$) or *S. mansoni* ($n = 21$). IgG binding to the microarray was measured for each individual. Background corrected median fluorescence intensity (MFI) values resulting from IgG binding to the different glycan fractions present on the microarray (x-axis) can be visualized on the heatmap. Raw data is available in Supplementary data 7. The top graphs display the averaged MFI values obtained for all individuals belonging to the same group (i.e., *S. haematobium*-infected, *S. mansoni*-infected or non-endemic control). The type of fraction content is indicated between the graphs and the heatmap: Blanks (negative controls), *Schistosoma* crude antigens, *N*-glycans (PNGase F or PNGase A released), *O*-glycans or GSL glycans from either *S. haematobium* or *S. mansoni* cercariae, worms or eggs (treated with HF or native). The raw data used to generate the graphs are provided as a Source Data file.

were built on the latter, while core-2 was more abundant in the eggs (Fig. 6, Supplementary data 4), as reported also for *S. mansoni O*-glycans[9,44]. In both life stages, HexNAc, Hex, and Fuc residues extended these cores, most likely forming LN and LeX motifs, as well as (fucosylated) terminal GlcNAc (Supplementary data 4, Fig. 6 and Fig. S8). MALDI-TOF-MS/MS analysis indicated the presence of isomeric structures, for instance in the eggs, where the ion species at *m/z* 1780.92 ($F_2H_3N_3$) corresponded to two structures, one presumably built on the branched Galβ1-3(Galβ1-6)GalNAc core and a linear version containing multiple LeX motifs (Fig. S8.B). *S. haematobium O*-glycans also contain fucosylated LDN motifs (Fig. 6 and Supplementary data 4), as *S. mansoni*[9], although the presence of DF was exclusively observed in the cercariae for *S. haematobium* with the linear glycan at *m/z* 1720.90 ($F_4H_1N_3$) and the structure with *m/z* 2907.42 ($F_8H_1N_5$) (Fig. 6, Fig. S8.B). The latter corresponds to the highly fucosylated core-2 *O*-glycan of composition $F_8H_1N_5$, also reported in *S. mansoni* eggs and cercarial glycocalyx[45]. The proportion of short, truncated *O*-glycans with *m/z* < 1100 was scarcer in *S. haematobium* eggs than in *S. mansoni*.

II] Differential recognition of *Schistosoma* glycans by antibodies from infected children

Knowing that many *S. mansoni* glycans are antigenic, we constructed microarrays of native glycans isolated from *S. mansoni* and *S. haematobium* to comparatively assess the antigenicity of *S. haematobium* glycans. Glycan-containing fractions were obtained through UHPLC-purification (as detailed in "Methods" and in Fig. S9) and selected to provide a broad coverage of the structural diversity of glycans from both species (data not shown), with a particular focus on GSL glycans among which major differences between *S. haematobium* and *S. mansoni* were observed. Details of glycan fraction compositions are provided in Supplementary data 5. A selection of monoclonal antibodies (mAbs) binding specific glycan motifs was used to ensure the printing quality of the generated microarrays (Fig. S9, Supplementary data 6) prior to incubation with infected and control sera. In addition, binding patterns of the mAbs confirmed the structural assignment of terminal LeX, LDN, and F-LDN(-F) motifs in *S. haematobium* GSL and *N*-linked glycans (Supplementary Data S2-3).

A) Immunogenic glycan motifs in *S. haematobium* and *S. mansoni* infections

Once validated, the glycan microarrays were screened for IgG and IgM binding with sera from children infected with either *S. haematobium* ($n = 18$) or *S. mansoni* ($n = 21$) and with sera from non-endemic individuals ($n = 9$). IgG (Fig. 7) and IgM (Fig. S10) responses yielded significantly higher median fluorescence intensity (MFI) values for both groups of infected children compared to the uninfected controls, with significantly higher binding to 117 (IgG) and 120 (IgM) array fractions out of 151 for the *S. haematobium*-infected group compared to the non-endemic control group (Supplementary data 7 and Supplementary data 8).

Infected individuals—with either of the two species—showed IgG and IgM responses higher than the control group to *N*- and *O*-linked glycans harboring LDN-F, fucosylated GlcNAc, (multifucosylated) LDN, or high-mannosidic structures (Man8-9) (Fig. S11). Higher IgM binding than the control group was measured to LeX-containing *N*-glycans but we noted the absence of anti-LeX IgG reactivity (fractions NG-7 and NG-

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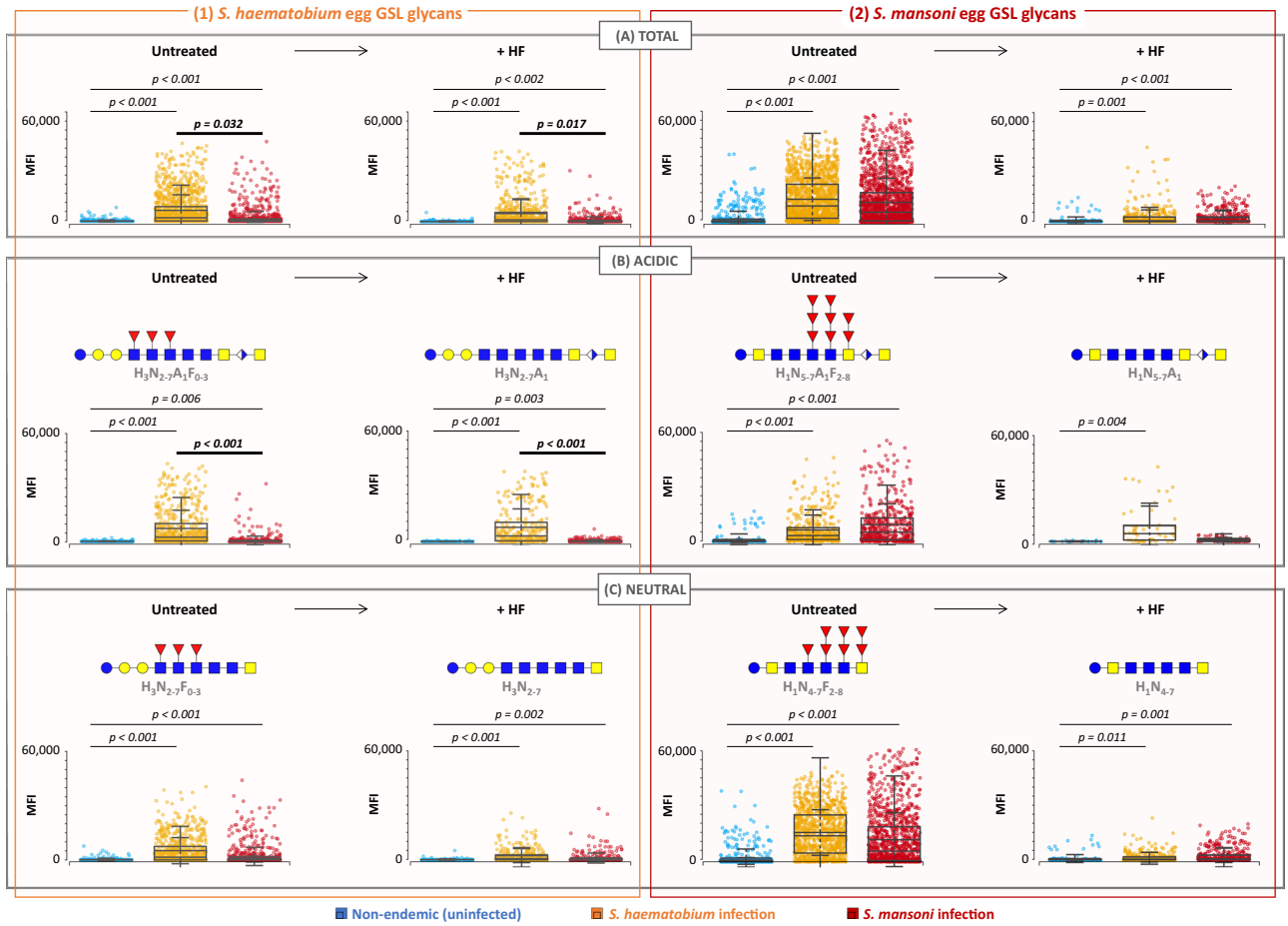


Fig. 8 | Differences in IgG binding to egg GSL glycans from *S. haematobium* and *S. mansoni* infection sera. Fractions containing egg GSL glycans were classified based on the species they originate from (*S. haematobium*/*S. mansoni*) and according to their structural characteristics: (**A**) total (all fractions included), (**B**) acidic (glycans containing GlcA in their composition) or (**C**) neutral (absence of GlcA in glycan compositions). A distinction was made between native (untreated) and HF-defucosylated (+ HF) glycan fractions. Glycans present in the fractions are shown above the boxplot charts, represented by a structure arbitrarily picked to illustrate the main structural features in each category. Compositions are indicated (H = Hex, N= HexNAc, A = GlcA, F = Fuc) together with a glycan structure, represented using the CFG nomenclature (blue square = GlcNAc, yellow square = GalNAc, red triangle = Fuc, blue circle = Glc, yellow circle = Gal, white and blue diamond = GlcA, or see symbol keys in Figs. 1–4). Each single dot in the bubble box plots corresponds to the MFI resulting from IgG binding to a particular fraction printed at a specific concentration for one individual (see Source Data). Boxplots (with median as center, 25th–75th percentiles (IQR) as box and whiskers extending to values within 1.5×IQR) show the IgG responses to the clustered glycan fractions for the three groups: non-endemic controls (blue), *S. haematobium*-infected (orange) and *S. mansoni*-infected (red). MFI values from fractions belonging to the same category were averaged for each individual. Statistical differences between *S. haematobium*-infected ($n = 18$), *S. mansoni*-infected ($n = 21$) and non-endemic ($n = 9$) individuals were determined using Bayesian statistics implemented in the *limma* R package. All tests were conducted as two-sided. P-values were adjusted for multiple comparisons using *limma*'s built-in procedures (Benjamini–Hochberg method), yielding adjusted false-discovery rates. Resulting p values (p) are indicated with connecting lines above the relevant boxplots. Exact p values and source data are provided as a Source Data file.

8, Fig. S11, Supplementary data 7 and Supplementary data 8), in line with previous reports in *S. mansoni* infections[13]. IgG and IgM binding to multifucosylated glycans yielded strikingly high MFI values in both infection groups compared to non-endemic sera (Fig. 7 and Fig. S10). Such structures harboring multifucosylated motifs were particularly abundant in *S. mansoni* GSL glycans and also present in some *O*-linked and GSL glycan-containing fractions isolated from *S. haematobium* cercariae and worms (Supplementary data 5), in line with our structural observations (Fig. 1). The few non-reactive fractions mainly contained truncated and mammalian-like *N*-glycans or short GSL glycans (Fig. 7 and Fig. S10, Supplementary data 5).

B) Differential IgG binding to acidic egg GSL glycans

Contrasting the above similarities, clear differences between the two infection groups were observed for IgG responses to fractions containing GSL glycans derived from *S. haematobium* eggs (Fig. 7). Significantly higher MFI values were measured to 24 fractions for IgG binding in *S. haematobium*-infection sera compared to *S. mansoni*-infection sera (Supplementary data 7). We further investigated IgG responses to egg GSL glycans by grouping fractions based on the species from which they originated with a distinction between intact glycans or glycans treated with HF for fucose removal (Fig. 8). Reflecting the observations made at the individual fraction level, the average of the MFI values measured for *S. haematobium* egg GSL glycans was significantly higher in *S. haematobium* than in *S. mansoni* infection sera (Fig. 8.A-1). When averaging MFI values recorded from IgG binding to *S. mansoni* egg GSL glycans (Fig. 8.A-2), on the other hand, there were no significant differences between the two infection groups ($p > 0.05$). HF-defucosylated *S. mansoni* fractions yielded markedly lower MFI values than those containing intact GSL glycans, showing that, for both infection groups, a substantial part of the IgG reactivity to *S. mansoni* egg GSL glycans was fucose-dependent. This was also observed for the IgG response from *S. mansoni*-infected individuals to *S. haematobium*-egg GSL glycans (Fig. 8.A-1). However,

this decrease in MFI levels between intact and HF-treated *S. haematobium*-GSL glycans was not as marked for *S. haematobium*-infected individuals, which also contain IgG against non-fucosylated epitopes.

Strikingly, most of the fractions to which differential IgG binding was observed between *S. haematobium*- and *S. mansoni*-infection sera were derived from *S. haematobium* eggs and contained acidic GSL glycans, i.e., structures that comprise GlcA in their compositions (Supplementary data 5 and Supplementary data 7). Therefore, we separately evaluated IgG binding to fractions based on acidic or neutral content, keeping the distinction between HF-treated and native glycans to assess the role of fucosylation in IgG binding (Fig. 8B, C). This analysis highlighted a strong IgG response to acidic GSL glycans in *S. haematobium*-infected individuals which was not observed in *S. mansoni*-infected individuals ($p < 0.001$, Fig. 8B-1). Glycan defucosylation by HF treatment had a very limited impact on IgG binding from *S. haematobium* infection sera, showing that the HF-resistant, GlcA-containing glycan epitope is a major IgG target in *S. haematobium* infections, but this was not the case for *S. mansoni* infections. *S. haematobium* and *S. mansoni* infection sera both showed a strong IgG binding to the acidic GSL glycans isolated from *S. mansoni* eggs, which was fucose-driven as a sharp decrease of IgG responses was observed to the HF-defucosylated versions of these glycans (Fig. 8B-2). Finally, there were no significant differences between both infection groups for IgG responses to the neutral GSL glycans ($p > 0.05$), confirming the role of the acidic residue in the preferential IgG binding from *S. haematobium* infections (Fig. 8C). Comparison of IgG binding from infection and control sera to the neutral fractions emphasized again the major role for fucosylated epitopes in IgG responses during schistosomiasis.

To summarize, the present glycan microarray-assisted study revealed that sera from *S. haematobium*-infected individuals contain IgG and IgM targeting a wide range of *Schistosoma*-derived glycans. These antibodies are partly directed to the same antigenic glycan elements as anti-glycan antibodies in *S. mansoni* infection sera. However, whereas fucosylation appears to be the dominant element in *S. mansoni*, the newly identified *Schistosoma* egg GSL glycans containing GlcA are highly antigenic and targeted by IgG responses specifically in *S. haematobium* infections.

Discussion

We report a comprehensive study of *S. haematobium* glycosylation and glycan antigens, a previously understudied aspect of this major schistosome species. We used a glycomic workflow of enzymatic and chemical glycan sequencing combined with MALDI-TOF-MS(/MS), UHPLC, and PGC-nano-LC-MS(/MS) analysis. The integration of various analytical methods proved effective in tackling inherent limitations of MALDI-TOF-MS-based workflows. Specifically, glycan sequencing provided compositional and linkage information while UHPLC allowed the study of short oligosaccharides with $m/z < 700$. Similarly, complex GSL glycan structures were resolved using PGC-nano-LC-MS/MS. This analysis revealed the extensive structural diversity and remarkable complexity of *S. haematobium* glycans, characterized by frequent isomeric variants and elaborate structures spanning a broad mass range. While complete structural elucidation of every individual glycan remains beyond the limits and scope of the present study, we were able to define the glycan repertoire of *S. haematobium* and uncover previously undescribed features of the GSL glycans in *S. mansoni*. Importantly, we identified several discriminative structural aspects between *S. haematobium* and *S. mansoni*.

Instead of the GalNAcβ1-4Glc disaccharide described in *S. mansoni*[33], *S. haematobium* GSL glycans are built on a core of three hexoses. With the presence of two putative ceramide glucosyltransferases in *S. haematobium* (MS3_00003792 and MS3_0000749)[28], orthologous to *S. mansoni*'s Smp_157080 and Smp_160210, we expect *S. haematobium* GSL glycans to be extended from glucosylceramide and thus the first ceramide-linked hexose to be Glc, as in *S. mansoni*[33]. We determined by UHPLC and exoglycosidase digestions that the ceramide-linked hexose was extended by a β1-4-linked Gal further elongated by an α-linked Gal (Fig. S4), with both α1-3 and α1-6 galactosyl linkages possible based on the specificities of the galactosidases used. PGC-nano-LC-MS analysis including GSL glycan standards indicated that the trisaccharidic GSL glycan core of *S. haematobium* is not identical to the globo-series core type-4 (Galα1-4Galβ1-4Glcβ1-1Cer) (Fig. S5.A), predominant in mammals[46], but corresponds to the isoglobo-series (Galα1-3Galβ1-4Glcβ1-Cer), previously identified in the GSL glycans of the liver fluke *Fasciola hepatica*[30,31]. To date, little is known regarding the enzymes involved in glycan biosynthesis in schistosomes, as many genes are predicted to code for a multitude of glycosyltransferases but very few have actually been studied and characterized in detail[27]. Thus, the glycan synthesis pathways in schistosomes are not well characterized[27] but the GSL glycan synthesis must differ between species. While *S. mansoni* uses a putative β1-4-N-acetylgalactosaminyltransferase to extend the glucosylceramide with GalNAc[33,47], *S. haematobium* employs both an α- and a β-galactosyltransferase to synthesize the trihexosyl core. Subsequent studies are needed to fully elucidate the structure of the *S. haematobium* GSL glycan core and its differential synthesis pathway compared to *S. mansoni*. Investigating the galactosyltransferases involved in the GSL glycan synthesis would also be valuable considering the structures containing branched β1-3 Gal, in addition to terminal β1-4 Gal, that were detected in *S. haematobium* cercariae (Fig. 1 and Supplementary data 2) but not in *S. mansoni*.

A recurring structural feature in both GSL and *N*-linked glycans throughout all life stages examined was the lower degree of fucosylation in *S. haematobium* compared to *S. mansoni*. We showed that highly fucosylated structures and triantennary *N*-glycans that are particularly abundant in *S. mansoni* mature eggs are not expressed by *S. haematobium* eggs (Fig. S2.D and Fig. S6.F). As they are presumed to originate from the egg-enclosed miracidia and the Von Lichtenberg's envelope in *S. mansoni*[9,48], it is possible that these differential glycans are related to the different excretion routes in the mammalian host and/or involved in specific schistosome-snail interactions.

Despite the lower degree of fucosylation of *S. haematobium* glycans, strong antibody responses to the multifucosylated structures printed on the microarray were observed in *S. haematobium* infection sera, similar to those observed during *S. mansoni* infections in the present work and in previous studies[13,41,42]. This indicates that the DF and TF motifs expressed by *S. haematobium*, mainly in cercarial *O*-linked and GSL glycans (Figs. 1, 6, Supplementary data 2 and Supplementary data 4) trigger substantial IgG and IgM responses in *S. haematobium* infections as well. Due to the uniqueness and importance of fucosylation in schistosomes, with respect to innate and adaptive immune responses in the mammalian host[8,11,13,14,42,49–52], the characterization of *Schistosoma* fucosyltransferases is of major interest. In silico studies have identified 14 full-length genes putatively encoding such enzymes in the genome of *S. mansoni*[53], and five of them have been functionally characterized in plants[54,55]. However, the enzyme(s) responsible for synthesis of the Fucα1-2Fuc linkage as in the DF and TF motifs remain unknown.

A major finding of the present study is the identification of acidic GSL glycans harboring GlcA-containing motifs, in adult worms and eggs. GlcA was previously found in schistosomes as part of the repeating unit [−6(GlcAβ1-3)GalNAcβ1-] of the *O*-linked CAA[56,57], a proteoglycan structure produced in the schistosome gut. Although we have not determined the exact linkage, sensitivity to β-glucuronidase indicates that GlcA is also β-linked in the GSL glycans. It is worth noting, however, that the context of GlcA in adult worm and egg GSL glycans, i.e. as single residues in terminal or subterminal positions of the glycan epitope (Fig. 3), greatly differs from the context of GlcA in CAA, where multiple residues are polymerized together with GalNAc as

a disaccharide repeating unit. GlcA is not a common substituent of mammalian *N*-linked or GSL glycans[58]. It is only observed in specific instances, as a rare building block in human plasma *N*-glycans where it can cap Gal residues in multiantennary structures[59,60] and as part of the sulfate (HSO_3)-containing human natural killer-1 (HNK-1) epitope (HSO_3-3GlcAβ1-3 Galβ1-4GlcNAc-R) which is found on a subset of human lymphocytes and in the nervous system[61]. On the other hand, GlcA appears to be a widespread monosaccharide in helminth glycans and has notably been reported in abundance extending LDN motifs in the *N*-linked and GSL glycans of parasitic nematodes where it forms antigenic epitopes[62–64]. Thus, it would be interesting to study its role in parasite biology and the interaction with the host, in addition to the observed antigenic properties. Perhaps capping of schistosome glycans by GlcA, as observed in the adult worm GSL glycans (Fig. 1, Figs. S2.B and S5.B) is a means of enabling anionic properties and electrostatic interactions that mimic those of the mammalian glycans often terminated with sialic acids[65]. On the other hand, GlcA-containing *N*-glycans have been found in molluscs[38]. Perhaps the intermediate snail host of *Schistosoma* also expresses GlcA-containing glycans which could indicate that GlcA is a factor in schistosome-snail compatibility.

In any case, GlcA clearly forms immunodominant glycan motifs in *Schistosoma* infections, here, in the context of the GSL glycans (Figs. 7–8, Fig. S10), and as part of the CAA[19]. Both individuals infected with *S. mansoni* and individuals infected with *S. haematobium* have serum antibody binding to CAA making this glycan a highly promising diagnostic target for pan-schistosome species serology. In the present work, we observed an IgM response to acidic GSL glycans in both *S. mansoni* as well as *S. haematobium* infection sera. However, significantly lower IgG binding was observed for *S. mansoni* infection sera in contrast to *S. haematobium* infection sera, that contain substantial serum IgG to acidic GSL glycans. We consider that these differences in IgG binding result from the lower abundance of GlcA-containing GSL glycans in *S. mansoni* and their heavy fucosylation. Indeed, not only the number of Fuc residues per glycan structure was much higher in *S. mansoni* eggs, but these residues were also mainly located terminally, and, thereby probably masking the GlcA residue, while the absence of dense fucosylation in its vicinity in *S. haematobium* creates an exposed, immunodominant GlcA-containing motif. Unraveling common and differential anti-glycan responses between the two species is of particular interest. Species-specific antigens could prove a useful tool to investigate important epidemiological questions since both *S. haematobium* and *S. mansoni* are present on the African continent[66], leading to areas of co-endemicity[67] and co-infections[68,69]. Moreover, improved detection of *S. haematobium* would help preventing serious and frequently misdiagnosed disease complications such as male and female genital schistosomiasis[20,21,70].

Further, differential glycosylation is an important consideration when investigating *Schistosoma* glycoproteins as potential vaccine candidates. In schistosomiasis, evidence of anti-glycan IgG exhibiting protective characteristics has been reported in several animal models, notably with IgG responses to fucosylated LDN motifs associated with infection and parasite clearance in macaques and baboons[14,51,52]. Given the variation observed in fucosylation levels in the present work, it appears important to validate glycan or glycoprotein vaccine candidates against the different *Schistosoma* species.

Accordingly, characterizing the protein and lipid carriers of the glycans identified in the present study remains a key challenge for future work, as it would provide valuable insights into the biological function of glycans. Determining which glycans modify excreted/secreted products is of great interest as such structures are likely to play major roles in host-parasite interactions. Although our glycan microarray-assisted study indicates that most glycan structures must be exposed, at least sufficiently to trigger antibody-responses from the host, it would be interesting to investigate their localization in further detail. Using immunofluorescence, previous work has shown highly specific surface-expression patterns of glycans in *S. mansoni* during cercarial development[71] and tissue-specific anti-glycan antibody responses have also been reported in *S. japonicum*-infected rats[72]. Such studies have not been performed for *S. haematobium*, and because the present work was conducted using homogenates of parasite life stages, it does not address the spatial localization of glycans. Recent advances in the field of MS imaging provide new avenues to study glycans in situ. Simultaneously, advances in analytical and computational tools combined with the expanded glycomic knowledge such as that provided here are now enabling comprehensive glycoproteomic studies[73].

While large-scale glycoproteomics are still lacking for schistosomes, targeted glycoproteomic analyses of excretory-secretory products from *S. mansoni* eggs have already contributed to major findings, notably with the characterization of the immunomodulatory glycoproteins Omega-1[49], Kappa-5[50], and Interleukin-4-inducing principle of *Schistosoma mansoni* eggs (IPSE)/α1[74]. Importantly, *S. mansoni* Omega-1 and IPSE/alpha-1 are internalized by monocyte-derived dendritic cells via the mannose receptor in a LeX-dependent manner[6,75] showing that besides triggering major antibody responses, *S. mansoni* glycans are involved in interactions with immune cells. LeX-terminated *N*-glycans were found to be significantly less expressed in *S. haematobium* than in *S. mansoni* cercariae (Fig. S7). Therefore, addressing whether *S. haematobium* and *S. mansoni* cercariae interact with different C type lectin receptors, leading to different innate immune responses, would be of interest. It has recently been reported that extracellular vesicles (EVs) from *S. mansoni* schistosomula and worms are differentially glycosylated and that the highly fucosylated schistosomula EVs are internalized by dendritic cell-specific intercellular adhesion molecule-3-Grabbing Non-integrin (DC-SIGN) while LDN-carrying adult worm EVs are recognized by the macrophage galactose-type lectin (MGL)[76,77]. Considering this, it would be valuable to investigate further the immune response to *S. haematobium* eggs as well. Fucosylated egg GSL glycans stimulate cytokine production in *S. mansoni* infections, while worm GSL glycans lacking LDN-DF do not[78]. Given that *S. haematobium* eggs do not express highly multi-fucosylated glycans, the probability that they exhibit different immune properties to those of *S. mansoni* is substantial.

To conclude, we provide an elaborate overview of the glycans expressed by different life stages of *S. haematobium* as well as insights into anti-glycan antibody responses during infection. The glycans expressed by *S. haematobium* share many features with those of *S. mansoni* but several structural specificities of *S. haematobium* glycans were also uncovered. These structural dissimilarities are reflected in differential IgG responses during infection with one *Schistosoma* species or the other, highlighting the importance of glycosylation in immunity. Our work contributes to a better understanding of *S. haematobium* (glyco)biology and paves the way to further research into the functional role of glycoconjugates in *S. haematobium* infections.

Methods

Structural characterization of glycans

Enzymatic release and AA labeling of N-glycans and GSL glycans from parasite proteins and lipids. The various life stages of *S. haematobium* (Egyptian NAMRU-3 strain)[78] and *S. mansoni* (Puerto Rican strain)[79] were obtained from the parasite life cycles maintained in male Golden Syrian hamsters (RjHan:AURA strain, Janvier labs, France) at Leiden University Medical Center in accordance with the Guide for the Care and Use of Laboratory Animals of the Institute for Laboratory Animal Research and have received approval from the Dutch 'Central Authority for Scientific Procedures on Animals' (CCD, License number: AVD11600202215904). Worms and eggs were recovered from hamsters 6–7-week post-infection for *S. mansoni* and 16–20-week post-infection for *S. haematobium*. *S. haematobium* and *S. mansoni*

cercariae were recovered from *Biomphalaria glabrata* and *Bulinus truncatus* snail shedding, respectively, at various time-points post-infection. About 1.2 million eggs, 125 000 cercariae and 100 worms from *S. haematobium* were used as starting material on two different occasions in order to generate biological replicates and about 0.8 million eggs, 150,000 cercariae and 240 worms were used from *S. mansoni* for direct comparison purposes. When unspecified, mixtures of eggs at different stages of maturity were used. To separate immature from mature eggs, a pool of 660,000 *S. haematobium* eggs was subjected to a Percoll gradient centrifugation protocol[80]. The eggs were first centrifuged in 40% Percoll in a saline solution of 1.7% sodium chloride (w/v). The immature egg layer was collected, and the pellet containing mature eggs was processed through a second centrifugation step in 50% Percoll prepared in saline to further eliminate residual debris. *N*-linked and GSL glycans were enzymatically released from ~100,000 of the mature and immature eggs thus isolated.

Glycoprotein/lipid preparation and glycan release were obtained through established protocols[9]. Pools of parasite material were ground in a Potter-Elvehjem homogenizer in 1 mL of ultrapure water (MQ, Milli-Q®, Millipore Sigma) and subjected to protein and lipid extraction by homogenization in methanol (MeOH) and chloroform in a final 4:7:13 ratio of MQ:MeOH:chloroform. Samples were sonicated and centrifuged at 1000 × *g* for 5 min, the upper phase removed and replaced by 50% MeOH.

This step was repeated twice and (glyco)lipids from the upper phase of extraction were subsequently purified using octadecylsilane (C18) cartridges (#7020-03 BAKERBOND® spe™, JT Baker®). Cartridges were conditioned with MeOH and 10% MeOH and samples were applied to the cartridges that were then washed with MQ. Flow-through and wash were collected, combined and used for an additional C18 purification on new C18 cartridges following the same procedure. Glycolipids were eluted from both cartridges with chloroform:MeOH:MQ(10:10:1) and elutions were combined and dried under nitrogen (N_2) flow. Dried (glyco)lipids were reconstituted in 500 μL 50 mM sodium acetate, pH5 containing 0.1% sodium taurodeoxycholate and lipid extracts were treated with 4 mU of recombinant rEGCase II (#4460, Takara) for 48 h at 37 °C. After 24 h, another 4 mU of rEGCase II was added. Following enzymatic release, GSL glycans were purified sequentially on C18 and carbon cartridges (#57088 Millipore Sigma Supelclean™ ENVI-Carb SPE). Released glycans were loaded on C18 cartridges conditioned with MeOH and MQ. Flowthroughs from load and from washes with 10% acetonitrile (ACN) and MQ were combined for each sample. The collected flowthroughs were next loaded on carbon cartridges conditioned sequentially with ACN, 50% ACN containing 0.1% trifluoroacetic acid (TFA) and MQ. After washing with MQ, the glycans were eluted with subsequent addition of 25% ACN then 50% ACN containing 0.1% TFA. Both elutions were pooled and dried in a SpeedVac.

Simultaneously, (glyco)proteins in the lower phase from the protein/lipid separation procedure described above were pelleted using excess volumes of MeOH. Protein pellets were dried under N_2 flow and then homogenized in PBS with 1.3% SDS and 0.1% β-mercaptoethanol. Denaturation was performed for 10 min at 95 °C, samples were cooled to room temperature and Nonidet P-40 (1.3% final concentration) added to each sample. To generate tryptic peptides, 100 μL of trypsin-coated Sepharose beads prepared according to manufacturer's instructions (GE Healthcare) were added to each sample and incubated overnight at 37°C. The beads were spun down, and peptide-containing supernatants collected. *N*-linked glycans were first released using 3500 units of PNGase F (#P0709, New England Biolabs (NEB)) and incubation at 37 °C for 24 h. Released glycans were cleaned up using C18 and carbon cartridges. Samples were loaded on C18 columns conditioned with ACN, 60% ACN containing 0.1% TFA and MQ and collected in the flowthrough of 10% ACN and MQ washes. Peptides were eluted from C18 columns using first 30% ACN with 0.1% TFA then 60% ACN with 0.1% TFA. Released glycans were further purified using carbon SPE as described for GSL glycans. Eluates from both C18 and carbon cartridges were dried in a SpeedVac. Peptide-containing C18 eluates were redissolved by sonication in 50 mM sodium acetate, pH 6 at 25 °C (Glycobuffer 3, NEB). 25 units of PNGase A (#P0707, NEB) were added to each sample prior to incubation for 48 h at 37°C to release potential PNGase A specific *N*-glycans. Reactions were cleaned up using C18 and carbon cartridges purifications sequentially as described above. C18 and carbon column eluates were dried in a SpeedVac.

Released and dried glycans were resuspended in MQ and labeled with AA by reductive amination with sodium cyanoborohydride as described[81]. Labeling was performed for 2 h at 65 °C. To remove excess labeling reagent, ACN was added to a final concentration of 75% and the sample was loaded onto Bio-Gel P10 Gel resin (#1504144, Bio-Rad) previously conditioned with MQ and 80% ACN. Resin was washed with excess volume of 80% ACN, then glycans were eluted with MQ and dried using a SpeedVac.

Separation of acidic and neutral GSL glycans for in-depth glycomic studies

About 1.8 million *S. haematobium* eggs, 365 *S. haematobium* worms and 2.5 million *S. mansoni* eggs were homogenized using a Potter-Elvehjem. Lipid extraction and enzymatic glycan release using rEGCase II was performed as detailed earlier. Subsequent C18 and carbon purifications were also identical except that the two final elution steps from the carbon cartridges (25% ACN and 50% ACN containing 0.1% TFA) were kept separate, resulting in two fractions for each sample, containing "neutral" and "acidic" GSL glycans, respectively. These fractions were dried in a SpeedVac and reconstituted in MQ. A fifth of each sample was aliquoted for PGC-nano-LC-MS/MS analysis (see below) while the rest was labeled with AA, as described above, for MALDI-TOF-MS analysis.

Chemical release and permethylation of O-glycans

About 200,000 eggs, 20,000 cercariae and 100 worms from the *S. haematobium* life cycle were subjected to reductive β-elimination to release *O*-linked glycans. Biological material from each stage was ground in a Potter-Elvehjem homogenizer in 4 mL MQ and lyophilized by freeze-drying. The dried samples were redissolved by sonication in 200 μL MQ, then split into two aliquots of 100 μL to generate technical replicates. 100 μL of a solution containing sodium borohydride (2 M) and sodium hydroxide (0.2 M) was added to each replicate and incubated at 40 °C for 24 h. Samples were neutralized on ice by adding 4 M acetic acid. MeOH with 1% acetic acid was added repeatedly with intermediate and final drying under a stream of N_2 until a neutral pH was reached. Released glycans were dissolved in 1 mL MQ and C18 purification identical to the *N*-glycan procedure was performed. C18 flowthroughs were subjected to sequential SPE using carbon cartridges previously conditioned in 80% ACN with 0.1% TFA and MQ After washing with MQ, the glycans were eluted with 25% ACN and 25% ACN with 0.05% TFA prior to concentration to dryness in a glass tube by N_2 flow. To facilitate MALDI-TOF-MS analyses *O*-glycans were next permethylated according to[82]. For this, 100 μL of DMSO saturated with sodium hydroxide powder was added to each tube and left for 10 min at room temperature. Next, 100 μL of iodomethane was added followed by an additional 10 min incubation at room temperature. Samples were then mixed by inversion in 400 μL of dichloromethane and 500 μL MQ. The aqueous layer was discarded and replaced with 500 μL MQ. This step was repeated at least five times or until neutral pH was reached. Finally, samples were dried under N_2 flow and redissolved in 80% MeOH for MALDI-TOF-MS analysis.

Glycan sequencing using hydrofluoric acid (HF) and exoglycosidase treatments

In order to elucidate glycan structures, released *N*-linked and GSL glycans were subjected to selective digestion of glycans. HF was used to remove α1-2/3/4-linked fucoses from the glycan backbone by mixing small aliquots of AA-labeled glycans suspended in MQ with cold 48% HF in a 1:100 ratio. Samples were incubated at 4 °C for 48 h and HF was then removed by evaporation under N_2 flow. Multiple washes were performed by addition and subsequent N_2 evaporation of MeOH. Finally, the HF-treated glycan samples were re-dissolved in MQ. A panel of exoglycosidases listed in Fig. 2 was used to selectively remove monosaccharides from glycans. Digestions of a few μL of glycan samples were performed overnight in recommended conditions. All enzymes were from NEB and used at 37 °C except for the β-glucuronidase (Millipore Sigma) used at 50 °C. Digestions with β-glucuronidase were performed in citrate buffer (100 mM, pH4) from Teknova (#Q2444) while reactions with α1-2,3,6 mannosidase and β1-3,4 galactosidase were performed in NEB glycobuffer 4 (50 mM sodium acetate, pH 4.5) supplemented with zinc for the α1-2,3,6 mannosidase. All other enzyme reactions were performed in NEB glycobuffer 1 (5 mM $CaCl_2$, 50 mM sodium acetate, pH 5.5), supplemented with BSA (1 mg/mL) for the α-galactosidases. Following HF and exoglycosidase treatment, clean-up and enzyme removal was performed using a previously detailed C18 clean-up[83]. Briefly, Millipore® Zip-Tips (Millipore Sigma # ZTC18S) were conditioned with 50% ACN containing 0.1% TFA and then with 0.1% TFA. AA-labeled glycans were bound to the C18 resin prior washing with 0.1% TFA. Finally, undigested controls and digestion products were eluted in 50% ACN containing 0.1% TFA.

MALDI-TOF-MS and MS/MS analysis

AA-labeled glycans. AA-labeled glycans released from different parasite life cycle stages and products of HF and exoglycosidase digestions were analysed by MALDI-TOF-MS using Bruker Daltonics mass spectrometers controlled by the software FlexControl 3.4 Build 119 (Bruker Daltonics). UltrafleXtreme® and Autoflex® instruments equipped with 1 kHz Smartbeam lasers and a Rapiflex® equipped with a 10 kHz Smartbeam-3D laser technology were used based on availability. Samples were spotted on a 384 well steel polished target plate. AA-labeled glycans solubilized in MQ were mixed on the plate with 2,5-dihydroxybenzoic acid (DHB) matrix (#8201346, Bruker Daltonics, 20 mg/ml in 30% ACN) while products of exoglycosidase digestions were directly eluted onto the plate in 50% ACN, 0.1% TFA mixed with DHB (10 mg/ml) at the end of the enzyme removal with C18 Millipore® Zip-Tips. All spectra were obtained in the negative-ion reflectron mode using Bruker® peptide calibration mix (#8206195, Bruker Daltonics) for external calibration. Spectra were obtained over a mass window of *m/z* 700 – 3500 with ion suppression below *m/z* 700 for a minimum of 20,000 shots (2000 Hz) obtained by manual selection of "sweet spots". The software FlexAnalysis (Version 3.4, Build 50, Bruker Daltonics) was used for data processing including smoothing of the spectra (Savitzky Golay algorithm, peak width: *m/z* 0.06, 1 cycle), baseline subtraction (Tophat algorithm), and manual peak picking. Peaks with a signal-to-noise ratio <4 were excluded as well as known non-glycan peaks such as glucose polymers. Deprotonated masses of the selected peaks were assigned using the GlycoPeakfinder® tool of the free software GlycoWorkBench[84] (Version 3, 29 June 2007). The AA label was taken into account as a fixed reducing-end modification and possible glycan composition was based on literature data on *S. mansoni*[9] and general knowledge of helminth glycobiology (i.e., 0–15 deoxyhexoses, 0–5 hexuronic acids, 0-20 hexoses and N-acetylhexosamines, 0–1 pentose). A deviation of 300 ppm was allowed for initial assignment of compositions. MS/MS was performed for structural elucidation via fragmentation ion analysis by MALDI-TOF/TOF on selected ions using the UltrafleXtreme® or Rapiflex® mass spectrometers in negative-ion mode. Information obtained from MS/MS profiles and from the various glycan sequencing procedures (i.e., HF treatment and exoglycosidase digestions) were combined to determine the final glycan composition associated with the detected *m/z* values in the MALDI-TOF-MS spectra and to characterize the corresponding glycan structures.

Permethylated *O*-glycans

Reduced and permethylated *O*-glycans (1 μL) were spotted onto a 384 well steel polished target plate with 1 μL DHB (10 mg/ml in 80% MeOH). They were analysed by MALDI-TOF-MS, similarly to the *N*-linked and GSL glycans, but using the UltrafleXtreme® instrument in positive-ion reflectron mode. MALDI-TOF-MS/MS fragmentation of selected ions was also performed in positive-ion reflectron mode and results of this analysis were used to assign the *O*-glycan structures.

Structural characterization of GSL glycans using PGC-nano-LC-MS and MS/MS

GSL glycans isolated from eggs of *S. haematobium* and *S. mansoni*, and from *S. haematobium* worms with the acidic and neutral fractions processed separately (see above) were next reduced and desalted by PGC SPE based on previously established procedures[85,86]. Briefly, GSL-glycans were reduced to alditols in 20 μL of sodium borohydride (500 mM) in potassium hydroxide (50 mM) for 2 h at 50 °C. Subsequently, 2 μL of 100% glacial acetic acid was added to neutralize the solution and quench the reaction prior to desalting by cation-exchange[87]. Glycans were eluted in MQ and dried by Speedvac. Dried samples were reconstituted in 0.1% TFA and loaded onto tips packed with 60 μL of bulk sorbent carbograph slurry (50 mg/mL in MeOH) previously conditioned three times with 100 μL of 80% ACN containing 0.1% TFA and three times with 100 μL of 0.1% TFA. Samples were washed three times with 50 μL of 0.1% TFA and eluted from tips twice with 30 μL of 60% ACN containing 0.1% TFA. Collected GSL glycan samples were dried by SpeedVac and reconstituted in MQ. Aliquots of released and reduced GSL glycans were treated with HF as described above. The defucosylated neutral GSL glycans from *S. haematobium* eggs were additionally subjected to digestion with β-HexNAcase using the reaction conditions detailed above. The GSL glycan samples resulting from HF and exoglycosidase treatments were cleaned again using C18 cartridges followed by PGC SPE as described above. After clean-up, reduced GSL glycans – native or HF treated/digested – were dried in a SpeedVac and re-suspended in 10 μL MQ and 2 μL were injected for PGC-nano-LC-MS/MS analysis. GSL glycan standards Gb3 and iGb3 were obtained by releasing the glycan moiety of synthetic trihexosylceramides from Avanti Research (Gb3-Ceramide #860699 and iGb3-Ceramide #860722) prior to reduction and clean-up as described above. A Dionex Ultimate 3000 nano-LC system (Thermo Fisher Scientific) equipped with a home-packed PGC trap column (5 μm Hypercarb, 320 μm x 30 mm) and a home-packed PGC nano-column (3 μm Hypercarb 100 μm x 150 mm) connected to an amaZon speed ETD ion trap (Bruker Daltonics) were used for the detection of glycan alditols. Mobile phase B contained 60% ACN and 10 mM Ammonium Bicarbonate Buffer (ABC), while mobile phase A was 10 mM ABC. To analyze GSL glycans, a gradient of B (1-51% at 0.5%/min) at a flow rate of 0.6 μL/min was employed. The temperature of the column was maintained constant at 45 °C. Ionization was achieved using the CaptiveSpray nanoBooster source (Bruker Daltonics) with a capillary voltage of 1000 V, dry gas temperature of 280 °C at 5 L min^{-1} and nebulizer at 3 pound per square inch (psi). Isopropanol-enriched dopant N_2 was used. MS spectra were acquired within an *m/z* range of 340–1850, smart parameter setting was set to *m/z* 800. MS/MS spectra were generated using collision-induced dissociation over an *m/z* range from 100 to 2500 on the top three most abundant precursors. Data were analyzed in Compass DataAnalysis, Version 5.0, Build 203 (Bruker Daltonics). Glycan structures were assigned on the basis of the known

MS/MS fragmentation patterns in negative-ion mode, elution order, and general glycobiological knowledge, with the help of GlycoWorkBench[84] (Version 3, 29 June 2007) as detailed above.

Raw MS and MS/MS data generated during this study have been deposited in Glycopost[88] (https://glycopost.glycosmos.org/, project ID: GPST000602).

Ultra-performance liquid chromatography (UHPLC) and mass spectrometry analysis of GSL glycans

AA-labeled GSL glycans isolated from *S. haematobium* eggs, HF-treated and digested with various exoglycosidases as described above were analyzed using UHPLC. UPHLC profiling was performed as described previously[29] with slight adjustments. A Waters Acquity H-class instrument (Waters Corp., Milford, MA, USA) composed of a binary solvent manager, a sample manager, a fluorescence detector (excitation wavelength 250 nm; detection wavelength 425 nm) and a QDa mass detector (settings: positive mode; target sampling rate: 10 point/sec; gain: 1; capillary voltage: 1.5 kV; probe temperature: 600 °C) was used. Glycans were separated using an Acquity BEH Amide Column (130 Å, 1.7 μm, 2.1 mm×150 mm; Waters Corp.) with 50 mM ammonium formate, pH 4.4 as solvent A and ACN as solvent B. A linear gradient of 12 to 47% of solvent A was applied from 0 to 35 min; 47 to 70% solvent A from 35 to 35.5 min; 70% solvent A from 35.5 to 36.0 min; 70 to 12% solvent A from 36 to 36.5 min and 12% solvent A from 36.5 to 40 min. Samples were kept at 5 °C prior to injection, and the separation temperature was 40 °C. Data from both the fluorescence and the mass detectors was acquired, processed and analyzed using Empower 3 software (Waters Corp.).

Comparative glycan microarray

Glycan microarray construction. A microarray was prepared using native AA-labeled GSL and *N*-linked glycans isolated from *S. mansoni* and *S. haematobium* that were fractionated following established UHPLC-based procedures[13,15,63,89] detailed in Fig. S9. To provide a broad coverage of the structural diversity of the GSL glycans from both species, 12 fractions derived from *S. haematobium* cercariae, 10 fractions from *S. mansoni* cercariae, 25 fractions from *S. haematobium* eggs and 36 fractions from *S. mansoni* eggs were included in the array together with subsets of GSL glycans from both species that were treated with HF to remove α1-2/3-linked fucoses to create unsubstituted GSL glycan backbones (Supplementary data 5). In addition, 16 *N*-glycan fractions from cercariae, adult worms and eggs from *S. haematobium* and *S. mansoni* were included to represent the various *Schistosoma N*-glycans. As a reference, crude antigens from eggs (soluble egg antigen, SEA) and from cercariae (cercarial antigen, CA) were included for each species, and five fractions containing *O*-glycans derived from *S. mansoni* cercariae and eggs were generated using material obtained during a previous microarray study[15] (more information is provided in Fig. S9). Aliquots of 20 pmol of the samples listed above were dispensed into a 384-well V-bottom polypropylene plate (#784201, Greiner Bio-One). When sufficient material was available, larger aliquots of 60, 100, 200, and 600 pmol were included. The plate was evaporated in a vacuum centrifuge, and glycans were redissolved in 10 μL of 1× spotting buffer (Nexterion Spot, #1066029, Schott AG) containing 10% DMSO, creating a glycan library for microarray printing at a minimum concentration of 2 μM, with higher concentrations of 6, 10, 20, and 60 μM used when available. The crude antigens (CA and SEA) were loaded to the plate at a concentration of 10 μg/mL, while the *O*-glycan pools had variable concentration comprised between 0.08 and 1.04 μM as specified in Supplementary data 5. Nine negative controls lacking biological material and consisting solely of buffer were included on the plate. Thus, the generated glycan library was constituted of 288 samples that were printed in triplicate to epoxysilane-coated glass slides (Slide E, Nexterion #1066643, Schott AG), using the Microgrid 600 microarrayer (Genomic Solutions) as further detailed in Fig. S9.

Glycan microarray screening

Glycan microarray validation. Upon printing, the glycan microarray was incubated with four mAbs reactive with known glycan motifs for validation purposes (Fig. S9).

Infection and control sera

S. haematobium infected sera ($n = 18$) were obtained from the highly endemic area of Lambaréné, Gabon. Children were enrolled in the previously described study[90] during which consent for anonymized use of sera for further serological studies was given. This study was approved by the ethics committees of the University of Tübingen and the International Foundation of the Albert Schweitzer Hospital, Lambaréné, Gabon. Infection was assessed using microscopic egg counting following urine filtration. Sera from children infected with *S. mansoni* ($n = 21$) were collected during a different previous study conducted in the fishing community of Piida, located on the shore of Lake Albert, Uganda[91]. Details on informed consent given by the participants and sample collection have been detailed in the original study, which was approved by the Uganda National Council for Science and Technology and cleared by the Office of the President. The study also received support from the Cambridge Local Research Ethics Committee. Diagnosis of infection was performed using the Kato-Katz technique on stool samples. Finally, anonymized sera from 9 individuals from non-endemic area were obtained from Sanquin (Dutch blood bank) and were used as uninfected controls.

Binding assay

The comparative glycan microarray was screened with infected and control sera for IgG and IgM binding. The protocols for slide incubation, scanning and data handling have been reported previously[13,72]. Briefly, residual epoxides on the glycan array slides were blocked using a solution of 2% BSA and 50 mM ethanolamine in PBS. Slides were then incubated with serum diluted 1:100 in PBS with 1% BSA and 0.01% Tween 20. Anti-glycan IgG and IgM binding was detected using goat anti-human IgG (Fc-specific) Cy3-conjugate (Millipore Sigma #C-2571) and goat anti-human IgM (heavy chain) Alexa Fluor™ 647 (Invitrogen #A-21249) as secondary antibodies. Fluorescence was measured using a G2565BA scanner (Agilent Technologies) operated and controlled using Agilent's Feature Extraction Software (version 7.0) at 10 μm resolution using two laser channels at 532 nm and 633 nm and with the sensitivity level of the photomultiplier tube set at 10%. Image analysis was processed with GenePix Pro 7.0 software (Molecular Devices) according to published methods[92]. Raw output files were deposited to Github (https://github.com/lmcpetralia/Major-human-schistosome-species-express-different-glycans) and the acquired data were exported to Excel. For each glycan-containing fraction, fluorescence intensities of triplicates were averaged and corrected for background using average fluorescence intensity of "blank" spots—i.e. spots printed from wells containing spotting buffer only - as a baseline. Finally, GLycan Array Dashboard (GLAD)[93], version 3.1, from the glycotoolkit website (glycotoolkit.com/Tools/GLAD/) hosted by the National Center for Functional Glycomics, was used for data visualization, data mining, generation of heatmaps and boxplot graphs.

Statistical analysis

Background corrected fluorescence intensities were normalized using log2 transformation and all the subsequent statistical data analyses were conducted using the publicly available statistical programming language R (http://CRAN.R-project.org/, version 3.5.0). Significant differences between log2 normalized datasets were evaluated using the R/Bioconductor software package *limma*. This package, optimized

for statistical analysis of microarray data uses a linear model approach to analyze microarray experiments and empirical Bayesian methods to assess significant differences between groups[94,95].

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

MALDI-TOF–MS(/MS) and PGC-nano-LC-MS(/MS) data supporting our findings have been made available in Fig. 1, Figs. 4–6, Figs. S1–7 and Supplementary Data S1-4. In addition, all raw MS data generated in this study have been deposited in the Glycopost[88] database under project ID GPST000602 [https://glycopost.glycosmos.org/]. Glycan microarray screening output files generated in this study have been deposited in Github [https://github.com/lmcpetralia/Major-human-schistosome-species-express-different-glycans]. The processed microarray data are available as Supplementary data 7-8. Source data are provided with this paper.

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Author contributions

A. v. D. and C. H. H. conceptualized the study; L. M. C. P., A. v. D., J. M. F., and C. H. H. conceived the experiments; A. v. D., M. W., J. M. F., and C. H. H. provided experimental guidance and supervision. L. M. C. P., and D. L. N. performed the MALDI-TOF-MS-based glycomics; L. M. C. P., and T. Z. performed the PGC-nano-LC-MS analysis; L. M. C. P., D. L. N., A. O. K., and C. H. S. constructed the glycan microarrays; L. M. C. P., T. V. and A. O. K. performed the glycan microarray screening and data analysis. A. v. D., A. O. K., P. G. K., M. W., and C. H. H. provided resources, with A. O. K. and P. G. K. having respectively provided control and infection sera for microarray screening. L. M. C. P. curated and analyzed the data and wrote the original draft. L. M. C. P., A. v. D., T. Z., A. O. K., M. W., J. M. F., and C. H. H. reviewed and edited the manuscript; J. M. F. and C. H. H. acquired the fundings. All authors read and approved of the final manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to Cornelis H. Hokke.

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